HiResNet: A Low-to-High Deep Learning Framework for Updating Large-Scale Land Cover Inventory

Fariba Mohammadimanesh, Mohammad Marjani, and Masoud Mahdianpari, *Senior Member, IEEE*

***Abstract*—Large-scale land cover (LC) mapping is critical for monitoring Earth's surface and managing environmental changes. While machine learning (ML) and deep learning (DL) methods have significantly advanced LC mapping efforts, the reliance on high-quality training labels remains a major limitation due to their time-intensive and costly collection process. To address this challenge, we propose HiResNet, a novel DL framework that combines a multiscale feature extractor (MSFE) within convolutional neural networks (CNNs), a vision transformer (ViT) architecture, and a weakly-supervised loss function, to produce a 10 m LC map using an existing low-resolution (LR) 30 m LC map. In particular, this framework reduces the dependence on high-resolution (HR) training labels while generating HR (10 m) LC maps using Sentinel-2 imagery and LR 30 m labels. The capability of the proposed method is evaluated in the province of Ontario (ON), Canada, a region characterized by a diverse landscape and different LC types, covering approximately 1.07 million km^2. The results demonstrate a remarkable capability of the proposed model to enhance spatial resolution and achieve high classification accuracy, with an overall accuracy (OA) of 89% for large-scale LC classification. Compared to conventional methods and advanced DL architectures, such as Random Forest (RF), ViT, HRNet, SegFormer, DeepLabV3, and L2HNet, the proposed method exhibits a superior performance by handling class imbalances and extracting complex features. Furthermore, uncertainty estimation using Shannon entropy and class disagreement metrics demonstrates that the highest levels of uncertainty are associated with object edges, narrow features (e.g., rivers and roads), and complex LC classes like forests.**

***Index Terms*—Convolutional neural network (CNN), deep learning (DL), land cover (LC), large-scale, multiscale feature extractor (MSFE), vision transformer (ViT), weakly-supervised.**

I. Introduction

THE land surface constitutes approximately 29% of Earth's total surface area and is critical for both carbon emissions and their subsequent sequestration [1]. Primary land cover (LC) types include water bodies, soil, vegetation, and impervious surfaces such as urban infrastructure [2]. The spatial distribution of these LC types shapes ecosystems, influencing the habitats and survival of diverse flora and fauna [3]. Consequently, LC is recognized as a crucial variable for environmental observation and has been integrated into the European Space Agency (ESA) Climate Change Initiative (CCI) program [4]. LC Mapping is an essential preliminary step in assessing and monitoring the condition of the Earth's surface, as well as understanding ecosystem responses to environmental changes. It provides a valuable understanding of landscapes' spatial and qualitative characteristics and offers a foundation for monitoring human activities and natural processes. Furthermore, LC mapping indirectly tracks critical environmental issues, such as biodiversity loss, deforestation, soil degradation, desertification, and hydrological alterations [5].

In the context of LC mapping, numerous satellite-based remote sensing techniques have been developed and refined over the years, using a variety of multispectral sensors, such as Sentinel-2 [6] and Landsat [7]. These observations enable the acquisition of invaluable data from the Earth's surface, which is crucial for accurately classifying different LC types. One of the key advantages of remote sensing is its ability to collect data across diverse spatial scales, ranging from local and regional to continental scales, making it particularly useful for large-scale LC mapping [8]. Remote sensing techniques for LC mapping rely on distinct spectral signatures of various LC types, which allows separability, characterization, and classification.

Semantic segmentation, which is often called LC classification in remote sensing, involves assigning predefined labels to each pixel within an image [6]. This technique is fundamental for generating pixel-level classification maps [7]. Traditional methods for semantic segmentation include algorithms, such as minimum distance, maximum likelihood, and logistic regression [9]. Despite their vast applications in earlier research, their dependency on the input data distribution and their inferior capability to handle high-dimensional remote sensing data have significantly limited their use in recent studies [10].

To address the limitations of traditional methods, there has been a shift toward using machine learning (ML) and deep learning (DL) algorithms for LC mapping. These techniques have demonstrated remarkable performance when applied to

Received 28 January 2026; accepted 18 March 2026. Date of publication 1 April 2026; date of current version 4 May 2026. This work was supported by the Natural Resources Canada's Earth Observation for Cumulative Effects Project. *(Corresponding author: Fariba Mohammadimanesh.)*

Fariba Mohammadimanesh is with Canada Centre for Remote Sensing, Canada Centre for Mapping and Earth Observation, Natural Resources Canada, Ottawa, ON K1A 0E4, Canada (e-mail: fariba.mohammadimanesh@nrcan-rncan.gc.ca).

Mohammad Marjani and Masoud Mahdianpari are with the Department of Electrical and Computer Engineering, Memorial University of Newfoundland, St. John's, NL A1B 3X5, Canada (e-mail: mmarjani@mun.ca; m.mahdianpari@mun.ca).

Digital Object Identifier 10.1109/TGRS.2026.3676677

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:44:22 UTC from IEEE Xplore. Restrictions apply.

satellite data for different remote sensing tasks, particularly using feature fusion strategies in forested regions [11], mountainous terrains [12], and heterogeneous landscapes [13], often leveraging cloud-based platforms or ensemble learning schemes [14]. These studies have demonstrated strong performance under fully supervised conditions, where training labels are available at the same spatial resolution as the input imagery, and classification is performed directly at the target scale.

ML and DL models excel at utilizing the rich spectral information inherent in remote sensing imagery, enabling precise pixel-level classification. This shift has resulted in widespread adoption of ML methods over the past few decades, particularly for generating large-scale LC products at moderate spatial resolutions (e.g., 30 m). Several notable projects demonstrate the success of these approaches. For example, in 2017, the United Kingdom Centre for Ecology and Hydrology (UKCEH) developed a 25 m LC product for the entire U.K., using time-series satellite imagery and ML classification techniques [15]. Similarly, the United States Geological Survey (USGS) routinely updates 30 m LC maps for the conterminous United States (CONUSs) every 2–3 years by applying ML algorithms to Landsat data [16]. At the continental level, the North American Land Change Monitoring System (NALCMS) is a collaborative initiative between Canada [Canada Centre for Remote Sensing (CCRS)], the United States (USGS), and Mexico to produce and update a harmonized 30 m LC map for North America every 5 years [17]. Globally, significant advancements have also been achieved using ML methods with 30 m resolution data over the past decade [18].

Although various LC products have been generated at national and global scales, the 30 m spatial resolution offered by Landsat data could be insufficient for tasks requiring finer spatial details. Many applications, such as urban planning, agricultural monitoring, and habitat mapping, demand higher resolution data to extract spatial patterns and localize features [19]. New opportunities have been gained by Sentinel-2 satellite data, offering a higher spatial resolution (10 m), enhanced temporal resolution (5 days), and a broader spectral range (13 bands) compared to Landsat imagery (11 bands). These advantages make Sentinel-2 particularly well-suited for producing LC products with a higher spatial resolution and potentially improved accuracy. Consequently, there is a growing demand for LC products with a 10 m spatial resolution to meet the needs of global, national, and even regional or provincial-scale studies.

With the rapid advancements in DL architectures, such as convolutional neural networks (CNNs) and transformers, the applied methodologies for almost all remote sensing tasks, particularly LC mapping, have experienced a significant transformation [20]. These cutting-edge algorithms excel at extracting high-level hierarchical features from satellite imagery, eliminating the need for auxiliary data or human intervention [19], [21]. This capability has made them invaluable for *LC* mapping and other remote sensing applications. However, the adoption of DL methods comes with notable challenges, as most DL models rely heavily on large volumes of high-quality labeled data for training [5]. These labels are often generated through labor-intensive and time-consuming manual annotation processes tailored to specific tasks, severely constraining the methods' scalability and mapping coverage [1]. This bottleneck highlights the need for innovative approaches that reduce reliance on extensive manual labeling while maintaining mapping accuracy.

To address limitations related to training data, Robinson et al. [22] developed a super-resolution loss function to utilize 30 m resolution LC maps as auxiliary label sources. By applying this approach, they successfully generated 1 m resolution LC maps for the north Chesapeake region of the United States. However, despite its achievements, this method still relies on the availability of partial high-resolution (HR) labels during the training process, with the low-resolution (LR) labels serving as supplementary data to augment the training dataset. Similarly, Malkin et al. [23] proposed a super-resolution loss function based on the statistical relationship between HR and LR labels. While this approach provides a theoretical framework for upscaling, it is constrained by the prerequisite of having HR labels available for training. The dependency on HR training data limits the scalability of these methods to regions where such high-quality labeled datasets are sparse or unavailable, posing a significant challenge for their broader application in large-scale LC mapping efforts. Accordingly, Dong et al. [24] introduced an online noise correction strategy and incorporated it into their segmentation framework to generate a 3 m national-scale LC map of China without any training data. To address the effects of noisy samples, Li et al. [1] introduced a low-to-high network (L2HNet) framework based on a CNN architecture and adopted a weakly supervision strategy. This framework employs a specialized loss function alongside a multiblock CNN design to upscale LR labels (30 m) and generate HR LC maps. The proposed framework demonstrated to be successful for generating an HR map of general LC categories (i.e., four LC classes) in six administrative states located in the Chesapeake Bay watershed of the United States, as well as a detailed classification map (15 LC classes) of the state of Maryland, covering an area of about 33 873 km^2. Liu et al. [25] first employed a conditional random field (CRF) model to refine the initial LC labels with 30 m resolution, and then applied a class-conditional label correction method to adjust noisy labels during model training and to produce LC map of China with spatial resolution of 10 m. Most recently, Li et al. [26] introduced a Semantic-augmented and noise-tolerant Network (SantNet) for HR urban LC mapping using publicly available LC products as imperfect labels and tested their model for mapping all urban areas across the CONUS.

As demonstrated, recent advances have shown that addressing noise in training labels can substantially improve the accuracy of HR LC mapping derived from imperfect reference data [27]. Nevertheless, most existing approaches remain constrained by coarse classification schemes and limited scalability, often focusing on broad thematic categories or explicitly focused on urban LC mapping. Despite these promising developments, further advances are required to refine methodological robustness and enhance class-level detail, particularly

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:44:22 UTC from IEEE Xplore. Restrictions apply.

for heterogeneous landscapes characterized by high intraclass variability, interclass similarity, and unbalanced class representation.

One particular area of improvement is related to the application of transformers within the DL models, which has been less investigated for large-scale applications. In particular, the vision transformers (ViT) architecture represents a state-of-the-art DL model, illustrating promising results in natural language processing (NLP), image classification, and segmentation tasks, including LC mapping, due to its superior ability to model global spatial relationships. Recent studies have successfully implemented transformer-based architectures for wetland mapping [20], LC mapping [28], and water body extraction [29], highlighting the growing applicability and advantages of transformers in remote sensing applications. Unlike conventional CNNs, which rely on localized kernels for feature extraction, ViT employs a self-attention mechanism that extracts long-range dependencies across an entire image [30]. This capability enables ViT to model complex spatial heterogeneity and makes it ideal for LC mapping tasks where intricate patterns exist across various resolutions.

To address the limitations and fill the gaps in the existing literature, we propose HiResNet, a novel end-to-end hybrid DL framework for HR LC mapping that leverages the complementary strengths of CNNs and transformers. Our approach generates HR LC map at 10 m using only existing publicly available LR 30 m LC maps as training labels and freely available 10 m Sentinel-2 satellite imagery as input data, overcoming the need for labor-intensive HR training data annotations. The proposed approach introduces several innovations in terms of methodology, scalability, adaptability, and thematic detail, which are essential for advancing the use of remote sensing in practical LC mapping. Specifically, the main contributions of this study are summarized as follows.

1) *Hybrid Architecture:* HiResNet is designed as a dual-branch module that extracts global contextual features via transformers and local spatial features through a multiscale feature extractor (MSFE) convolutional block. The MSFE module is particularly effective for heterogeneous landscapes, where the scale variability of LC categories limits conventional CNNs.
2) *Resolution Discrepancy Compensation Mechanism:* A confidence area selection (CAS) module is adapted to guide the training process by distinguishing between high-confidence and uncertain regions, which informs the computation of loss. A novel HiRes loss, comprising weighted cross-entropy (WCE), feature consistency (FC), and intersection-over-union (IoU) components, is also proposed. These elements enhance pixel-level accuracy, maintain spatial consistency, and manage prediction uncertainty in complex LC mapping tasks.
3) *Extensive Experimental Validation:* We rigorously evaluate HiResNet through benchmark comparisons with both general-purpose semantic segmentation networks and specialized state-of-the-art noise-robust models. Detailed ablation studies quantify the contribution of each module. Results consistently demonstrate that HiResNet outperforms existing methods, confirming that the proposed modules collectively drive significant improvements in segmentation accuracy.
4) *Quantifying Uncertainty and the Impact of Low-Resolution Labels:* We systematically assess the uncertainty and implications of using LR labels for training DL models, employing metrics, such as Shannon entropy and class disagreement, to characterize prediction confidence across different LC categories.
5) *Expanding LC Granularity:* We incorporate a broader and more detailed LC categories than commonly used in prior works by applying the proposed method to a large-scale (~1.07 million km^2) heterogeneous landscape and thus, demonstrating practical scalability of our proposed model for the semantic segmentation task.

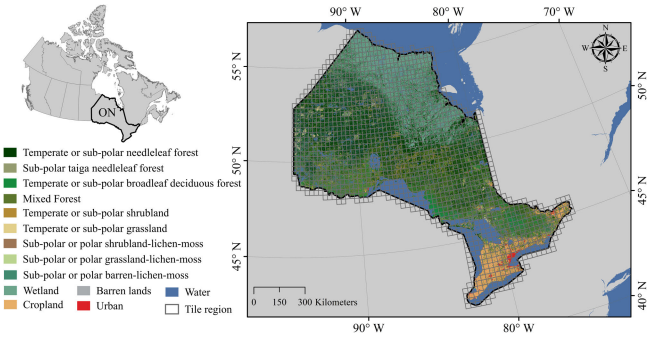


Fig. 1. Extent of the study area, which is divided into 1706 equal regions, shown as gray rectangles. The LC data, with a spatial resolution of 30 m, was obtained from CCRS Natural Resources Canada (NRCan) [32].

While the capability of state-of-the-art noise-robust DL models has been studied for LC classification in the United States and China, to the best of our knowledge, this is the first study to develop and evaluate such an advanced framework for LC mapping in Canada with a highly heterogeneous landscape and different LC categories than those examined in the previous studies.

II. Study Area and Dataset

A. Study Area

Ontario (ON), the second-largest province in Canada by land area and the most populous, spans approximately 1.07 million square kilometers and encompasses various landscapes. ON covers a diverse range of landscapes from the temperate regions along the Great Lakes to the boreal forests in the north [31]. LC types within the study area encompass a diverse range of classes, namely temperate or sub-polar needleleaf forest, sub-polar taiga needleleaf forest, temperate or sub-polar broadleaf deciduous forest, mixed forest, temperate or sub-polar shrubland, temperate or sub-polar grassland, sub-polar or polar shrubland-lichen-moss, sub-polar or polar grassland-lichen-moss, sub-polar or polar barren-lichen-moss, wetland, cropland, barren land, urban area, and water. Given the large size of the study area, it was divided into 1706 smaller regions, each approximately 22.1 km by 28.5 km in size. These smaller regions are referred to as tiles here, as illustrated in Fig. 1.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:44:22 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CLASS MERGING RELATIONS BETWEEN THE 30 M LC MAP AND THE NEW TARGETED CLASSES IN THIS STUDY

Original LR LC class	% of ON before merging	Modified LR LC class
Temperate or sub-polar needleleaf forest	31.1	Needleleaf Forests
Sub-polar taiga needleleaf forest	0.005	
Temperate or sub-polar broadleaf deciduous forest	5.14	(Broadleaf) deciduous forest
Mixed Forest	14.22	Mixed Forest
Temperate or sub-polar shrubland	2.90	Shrubland
Temperate or sub-polar grassland	0.00001	Grassland
Sub-polar or polar shrubland-lichen moss	2.19	Lichen and moss
Sub-polar or polar grassland-lichen moss	0.0001	
Sub-polar or polar barren-lichen-moss	0.16	
Wetland	19.85	Wetland
Cropland	4.71	Cropland
Barren land	0.07	Barren
Urban	1.27	Urban
Water	18.32	Water

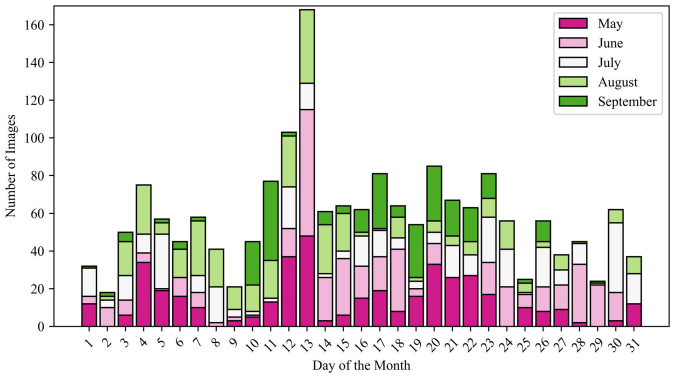


Fig. 2. Number of images used in this study from early May to end of September, categorized by day of the month.

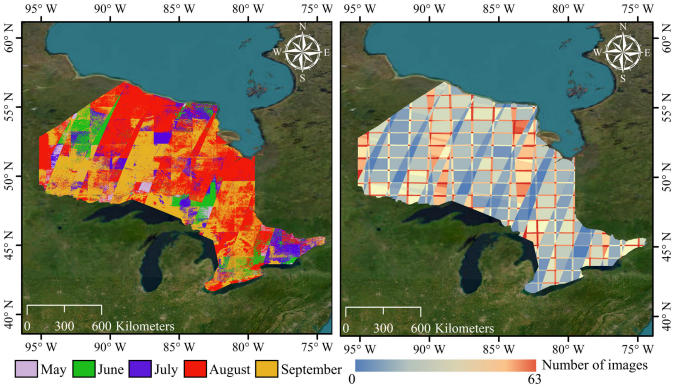


Fig. 3. Distribution of Sentinel-2 images used in the study across the study area. (Left) Temporal distribution of images used for data compositing—May (light purple), June (green), July (dark purple), August (red), and September (yellow). (Right) Spatial distribution shows the number of available images in ON between May to September 2020.

B. LR Labels

CCRS has produced a 30 m LC map of Canada using Landsat data, since 2010, and updated the product every five years. This dataset includes 14 LC classes in ON, compatible with the level II NALCMS classification system, as illustrated in Fig. 1. For this study, the CCRS 2020 LC map was used and clipped to the boundary of ON [32]. The area of ON covered by each class varies significantly; some classes, such as sub-polar taiga needleleaf forest, are rare, while others, like temperate or sub-polar needleleaf forest, dominate large portions of the study area. To address this imbalance issue to some extent, classes with very small areas were merged with similar classes. Table I presents both the original (i.e., the 2020 LC map categories and their percentage) and modified LC classes (i.e., this study) after merging the rare classes. The merging was performed based on the level I NALCMS classification system, featuring 12 LC categories. However, the class of snow/ice was removed in this research, as there is no permanent class of snow/ice in this area. Accordingly, the resulting LC map in this study has 11 LC classes.

C. Sentinel-2 Satellite Data

Sentinel-2 is a satellite mission of the ESA Copernicus program, designed to monitor land and vegetation. The Sentinel-2 constellation consists of two satellites, Sentinel-2A and Sentinel-2B, offering 13 spectral bands, ranging from visible to shortwave infrared (SWIR) spectrum, at spatial resolutions of 10, 20, and 60 m from different bands.

In this study, the Sentinel-2 satellite data were collected from Google Earth Engine (GEE) cloud platform for the period between early May and the end of September 2020, as this period typically corresponds to the growing season in the study area, making it ideal for the LC mapping. Fig. 2 illustrates the number of images used in this study, categorized by each month and the corresponding days within the month. Both 10 and 20 m bands from Sentinel-2 satellite data were used, with the 20 m bands resampled to 10 m resolution using the nearest neighbor resampling method. In addition, images with cloud coverage below 5% were selected to ensure data quality and reliability. Fig. 3 shows the distribution of images across each month and the number of available images within the study area.

In addition to the 10 and 20 m spectral bands of Sentinel-2 satellite data, five spectral indices were used as auxiliary bands: normalized difference vegetation index (NDVI), normalized difference built-up index (NDBI), modified normalized difference water index (MNDWI), green normalized difference vegetation index (GNDVI), and bare soil index (BSI). Each of these indices provides valuable discriminative information useful for large-scale LC mapping. For

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:44:22 UTC from IEEE Xplore. Restrictions apply.

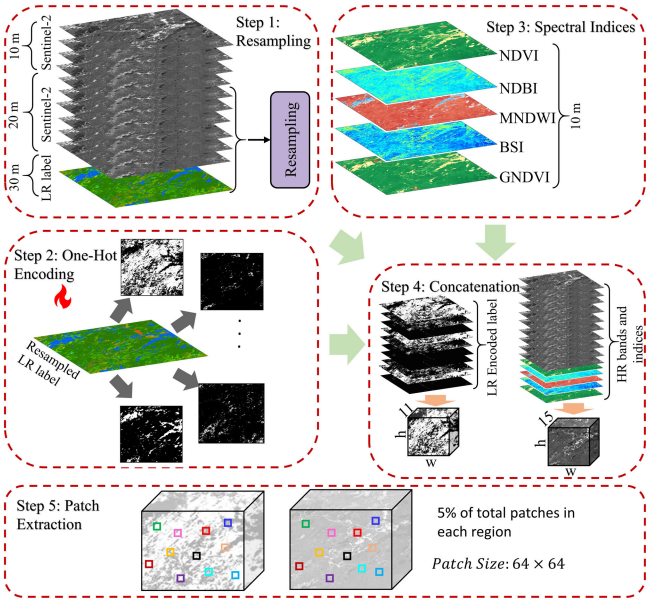


Fig. 4. Overview of the data preparation process for generating the 10 m LC map.

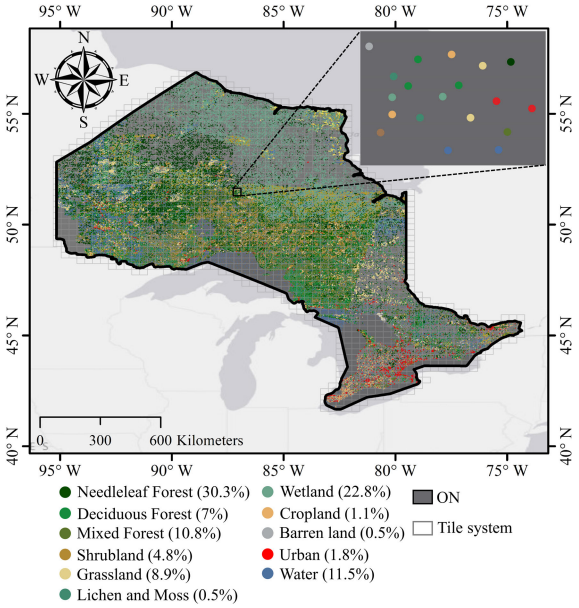


Fig. 5. Spatial distribution and class proportions of the approximately 150 000 manually annotated validation samples across 11 LC classes within ON. Percentages in the legend indicate each class's share of the total validation sample.

example, NDVI provides information about vegetation characteristics, NDBI highlights built-up areas, MNDWI identifies water bodies, GNDVI measures vegetation density and condition, and BSI characterizes bare soil and sparsely vegetated soil. These indices enhance the classification and analysis of LC by extracting specific features related to the landscape.

For each Sentinel-2 spectral band and derived spectral indices, multiple cloud-free observations acquired between May and September 2020 were temporally composited into a single representative value per pixel using a median compositing strategy. Median compositing was selected to reduce the influence of residual clouds, atmospheric variability, and short-term noise while preserving seasonal surface characteristics. The same compositing procedure was applied consistently to all spectral bands and indices before patch extraction.

D. Data Preparation

The preprocessing pipeline for generating 10 m LC maps from Sentinel-2 data using the 30 m LC map as training labels involves a standardized and reproducible sequence, consisting of: 1) spatial resampling; 2) label encoding; 3) spectral index computation; 4) feature concatenation; and 5) patch extraction, as illustrated in Fig. 4.

First, Sentinel-2 bands with a spatial resolution of 20 m are resampled to 10 m using the nearest neighbor method to align them with the higher resolution bands. Similarly, the LR LC map labels, initially at 30 m resolution, are resampled to 10 m to match the spatial resolution of the input data. Next, a one-hot encoding process is applied to the LR labels, where each LC class is represented as a separate binary band, with 1 representing the presence and 0 representing the absence of a specific class. This encoding facilitates compatibility with the DL framework by standardizing the label format.

In the third step, five spectral indices, including NDVI, NDBI, MNDWI, GNDVI, and BSI, are calculated using Sentinel-2 spectral bands. Next, the one-hot encoded LR labels, resampled Sentinel-2 bands, and calculated spectral indices are concatenated into a single multilayer data cube. This consolidated dataset combines all the features and inputs required for the DL model.

Finally, the data are divided into smaller patches of 64×64 pixels in the last step to prepare them for model training. To ensure a representative sample, 5% of each region's available patches are randomly selected. These patches serve as input data for training the proposed DL model, facilitating efficient processing and accurate generation of the HR 10 m LC map.

E. Ground Truth Data

In Section II-D, we described the extraction of 5% of all possible 64×64 patches within each region, with no overlap, which were used for the training process. For the validation process, a separate collection of sample points was gathered. These points were labeled using a combination of field data collected through on-site visits, as well as visual interpretation of HR aerial and satellite imagery, with the assistance of experts who were thoroughly familiar with the study area.

To ensure that the validation set was independent of the training data, we excluded any sample points that overlapped with the patches previously selected for training. The remaining data points that did not intersect with the training patches were then used for the validation process. Fig. 5 illustrates the spatial distribution of these validation points across the study area.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:44:22 UTC from IEEE Xplore. Restrictions apply.

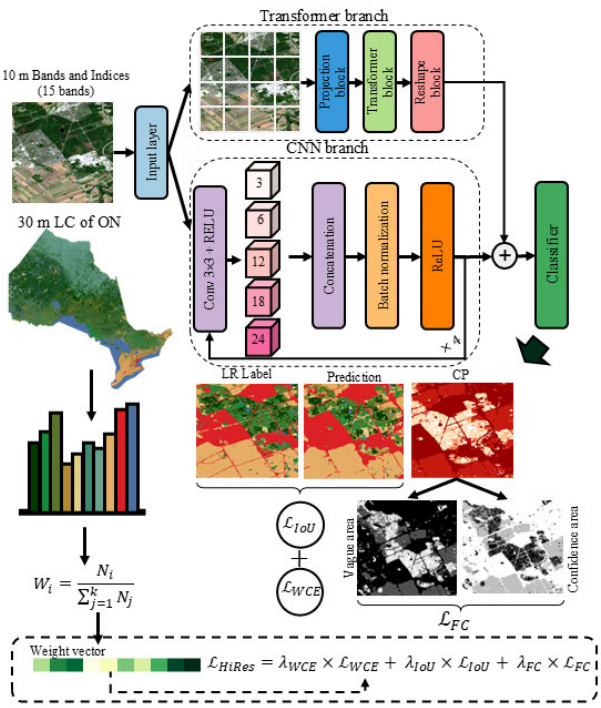


Fig. 6. Overall architecture of the proposed DL framework for generating the 10 m LC map from LR labels and Sentinel-2 data. The framework integrates two branches: a transformer branch (top) for extracting global contextual features through projection, transformer, and reshape blocks, and a MSFE branch (bottom) for extracting local spatial features using convolutional layers. The outputs from both branches are concatenated and passed to a novel weakly-supervised loss function.

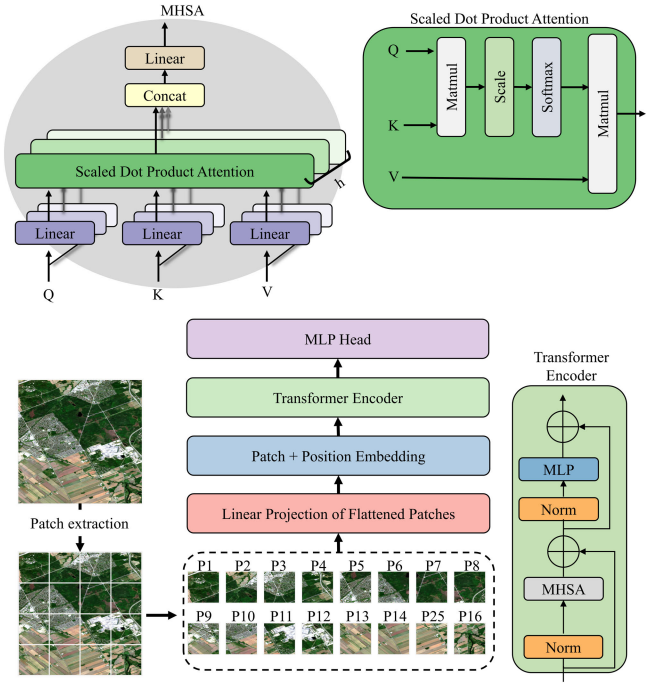


Fig. 7. Overview of ViT architecture used in this study.

III. Proposed Framework

The overall framework of the proposed model is depicted in Fig. 6. The model is designed as an end-to-end trainable DL architecture that uses 10 m Sentinel-2 satellite data and LR (30 m) LC labels to generate an HR (10 m) LC map at large-scale. Overall, HiResNet consists of two main components: 1) two branches of DL modules for extracting local and global features, and 2) a novel weakly-supervised loss function specifically designed and developed for refining LC predictions from LR labels. The details of these components are presented in the following sections.

A. ViT Branch

ViT architecture consists of five key components: 1) patch extraction; 2) linear projection; 3) positional embeddings; 4) a transformer encoder; and 5) a multilayer perceptron (MLP) head. As depicted in Fig. 7, these elements work together to process each input image. At first, ViT divides the input image $X \in R^{H\times W\times C}$, where H, W, and C, are the height, width, and number of spectral bands, respectively, into nonoverlapping patches of size $P \times P$. Each patch is flattened into a vector $x_p \in R^{P^2\times C}$. These vectors are then linearly projected into a D-dimensional embedding space using a learnable matrix E, producing patch embeddings $z_p = x_pE$, where $z_p \in R^D$ and D is the embedding dimension. To encode positional information, a learnable positional embedding $PE \in R^{N\times D}$ is added to each patch embedding, where $N = (H \times W)/(P^2)$ is the number of patches. The input sequence for the transformer becomes $Z_0 = \{z_1; z_2; \ldots; z_N\} + PE$. The transformer encoder comprises L stacked layers, each consisting of a multihead self-attention (MHSA) mechanism and a feed-forward network (FFN). The MHSA mechanism computes self-attention scores for each patch embedding by projecting them into query Q, key K, and value V spaces using learnable weight matrices W^Q, W^K, and W^V. The self-attention is calculated as follows:

$$Attention\,(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{D_k}}\right)V \tag{1}$$

where D_k is the dimension of the query and key vectors, ensuring numerical stability. The outputs of multiple attention heads are concatenated and linearly transformed. The FFN, composed of two fully connected layers with a nonlinear activation function [e.g., Gaussian error linear unit (GELU)], further refines the features [33]. Both MHSA and FFN are encapsulated with residual connections and layer normalization (Norm), ensuring stable training [34]. The final output from the transformer encoder is typically passed through a task-specific head to generate the final prediction. However, in this study, the transformer encoder's output is directed to a combination block for further processing.

B. MSFE Branch

The MSFE module is particularly useful for LC mapping tasks using Sentinel-2 imagery and LR 30 m LC training labels due to its ability to address the scale variability of various LC categories in satellite images. LC classes, such as water bodies, urban areas, vegetation, forests, and bare soil, exhibit features at different spatial scales. For example, urban areas often contain small-scale structures like buildings and roads, whereas forests occupy broader regions. Extracting both fine-grained local details and large-scale contextual information is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:44:22 UTC from IEEE Xplore. Restrictions apply.

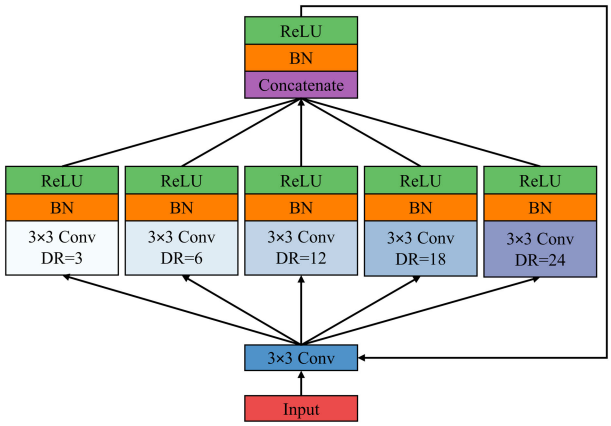


Fig. 8. MSFE module architecture. The module consists of 3 × 3 parallel convolutional layers (Conv) with varying DRs (DR = 3, 6, 12, 18, and 24). Each convolutional output experiences batch normalization (BN) and a rectified linear unit (ReLU) activation function before being concatenated along the channel dimension.

essential to accurately classify these diverse classes [35]. Fig. 8 indicates the MSFE module used in this study.

A parallel dilated CNN network is used with multiple dilation rates (DRs) to extract multiscale feature information. The dilated convolution operation expands the receptive field of the convolutional kernel without increasing the computational cost [36]. Mathematically, the dilated convolution for an input feature map $x(p)$ at position p with kernel W and DR d is defined as follows:

$$F_d(p) = \sum_{k \in K} W(k) . x(p + d.k) \tag{2}$$

where K represents the kernel size, and $p + d \cdot k$ denotes the offset for the DR d. Smaller DRs (e.g., $d = 3$) extract fine-grained details of LC classes, such as small urban features, while larger DRs (e.g., $d = 24$) are more suitable for capturing broader features, such as forests and water bodies. The outputs from parallel convolutions with varying DRs are combined through a concatenation layer along the channel dimension, expressed as follows:

$$F_{Concat} = \left[F_{d_1}, F_{d_2}, \ldots, F_{d_n}\right] \tag{3}$$

where F_{d_i} represents the feature map obtained from a specific DR d_i. This combination and the designed architecture enable the network to learn a rich, multiscale representation of the input data, extracting both localized and global patterns using the MSFE branch [37].

C. Combination Block

The output of the transformer encoder is first upsampled using transposed convolutions to achieve a spatial resolution of $W \times H \times 1$, matching the desired dimensions of the LC map. Simultaneously, the MSFE module processes the input features using dilated CNNs, and its output is passed through a CNN layer with a single filter to produce a feature map of the same shape $W \times H \times 1$. These two feature tensors, one containing global contextual information from the transformer encoder and the other emphasizing multiscale local details from the MSFE, are combined using an addition operation. This fusion of global and local features allows the integration of both coarse-level contextual information and fine-grained spatial details. Finally, a CNN layer with 11 filters (equal to the number of LC classes), followed by a Softmax activation function, is applied to the combined feature map to produce the final LC classification map.

D. Loss Function

The task of generating HR LC maps at a 10 m spatial resolution from Sentinel-2 images and LR 30 m ground-truth labels requires an advanced loss function that accommodates the challenges of HR image segmentation, uncertainty in predictions, and potential class imbalances. To address these issues, this study proposes HiRes loss, which is used in the training phase of the HiResNet model for generating 10 m LC maps from 30 m labels. The HiRes loss integrates several components: CAS, WCE loss, FC loss, and IoU loss. These components work synergistically within the HiResNet model to provide accurate pixel-level predictions, enhance spatial consistency, and handle prediction uncertainty.

1) CAS Module: The CAS module is a key in distinguishing between confidence and vague areas (VAs) in the predicted map. In many segmentation tasks, a model may need clarification about its predictions in certain regions. CAS enables the model to focus more on areas with high confidence and less on uncertain regions. The CAS process is visualized in Fig. 9. The maximum predicted probability across all classes is calculated for each pixel and called confidence probability (CP). If the highest probability exceeds a predefined threshold (τ), the pixel is considered to belong to the confidence area (CA); otherwise, it is classified as part of the VA area. The differentiation between CA and VA allows the loss function to act differently in these areas and thus, the model optimizes its performance in regions where it is most confident.

1) *CA:* Pixels where the model has high confidence in its class predictions.
2) *VA:* Pixels where the model has less confidence in its class predictions.

Accordingly, the CA and VA masks are generated and aid in adjusting the computation of loss, ensuring that uncertain predictions do not significantly affect the overall model performance.

2) FC Loss: The FC loss is designed to penalize significant differences between the feature representations of CA and VA areas. This loss encourages the HiResNet model to reduce VAs. The primary goal of this loss is to enforce consistency across the predicted map, ensuring the HiResNet model does not produce erratic predictions in uncertain regions where the ground truth is less clear. Mathematically, the loss computes the variance between the averaged predicted features for CA and VA as illustrated in the following equations:

$$\mathcal{L}_{FC} = \|F_{CA} - F_{VA}\|^2 \tag{4}$$

$$F_{CA} = \frac{\sum_{i,j} M_{CA}(i,j) P(i,j)}{\sum_{i,j} M_{CA}(i,j)} \tag{5}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:44:22 UTC from IEEE Xplore. Restrictions apply.

$$F_{VA} = \frac{\sum_{i,j} M_{VA}(i,j)\, P(i,j)}{\sum_{i,j} M_{VA}(i,j)} \tag{6}$$

where M_{CA} is the mask for the CA and $M_{CA}(i,j) = 1, M_{VA}$ is VA mask, where $M_{VA}(i,j) = 1 - M_{CA}(i,j)$, P is prediction made by HiResNet model, F_{CA} is pooled predictions for CA, and F_{VA} is pooled predictions for VA. The FC loss helps to maintain the spatial consistency of the model's predictions by penalizing feature disparities between regions with high CA and those with uncertainty (VA). This module improves the model's ability to handle ambiguity while preventing sharp discontinuities in the predicted map.

3) WCE Loss: The WCE loss minimizes the discrepancy between the predicted probability distribution and the actual label distribution. This loss is applied only to the CA, allowing the model to focus on improving prediction accuracy in regions where it has high confidence. To address the challenge of class imbalance, common in LC mapping, where certain classes (e.g., lichen and moss or barren) may be underrepresented, the cross-entropy (CE) loss is weighted using class-specific weights. These weights are calculated based on the ratio of each class to the total area in the entire study area to ensure the importance of each class is appropriately adjusted. Underrepresented classes are given higher weights, which helps prevent the model from becoming biased toward over-represented classes. The final WCE loss is computed as the average of the pixel-wise losses in the CA regions, guiding the model to optimize for accurate predictions in regions where it is most certain. Mathematically, WCE is calculated using the following equation:

$$\mathcal{L}_{WCE} = \frac{1}{N_{CA}} \sum_{i=1}^{B} \sum_{j=1}^{H} \sum_{k=1}^{W} M_{CA}(j,k) \sum_{c=1}^{C} W_c \times Y_T^{(i,j,k,c)} \times \log\left(P^{(i,j,k,c}\right) \tag{7}$$

where N_{CA} is the number of confidence pixels, Y_T is one-hot encoded ground truth labels with the shape of [batch size (B), height (H), width (W), and number of classes (C)], W is class weights, a vector of size C, and M_{CA} is CA mask, where $M_{CA}(i,j) = 1$ for confidence pixels and zero otherwise.

4) IoU Loss: The IoU loss is a metric used to evaluate the quality of pixel-wise segmentation predictions in HiResNet. Although in this article, LR labels have been used for the training process, it is assumed that the LR labels possess a certain level of accuracy (which is further investigated in the results section), and this level of correctness is leveraged in the HiResNet model. Despite the potential discrepancies between the predicted and actual labels, the IoU loss still plays a valuable role by reinforcing spatial alignment in confident regions. IoU quantifies the overlap between the predicted and actual label masks for each class, aiming to maximize the IoU and improve the spatial accuracy of the predictions. However, to facilitate optimization, we use $1 - IoU$ as the loss function [see (8)], where the objective is to minimize this value. IoU is computed as the ratio of the intersection (the common pixels between the predicted and ground-truth classes) to the union (all pixels that belong to either the predicted or true classes). To prevent issues with empty intersections or unions, which could lead to division by zero, a small epsilon value ($\varepsilon = 1e{-}7$) is added. By incorporating $1 - IoU$ into the HiResNet loss function, the HiResNet model produces accurate pixel-level classifications and enhances the spatial coherence of the LC map, making the predictions more consistent and precise across the entire region.

$$\mathcal{L}_{IoU} = 1 - \frac{\sum_{i,j,k}(Y_T \cap Y_P)}{\sum_{i,j,k}(Y_T \cup Y_P) + \varepsilon}. \tag{8}$$

5) HiRes Loss: The HiRes loss is a weighted sum of the WCE, FC, and IoU losses. WCE, IoU, and FC losses are scaled by hyperparameters ($\lambda_{WCE}, \lambda_{IoU}$, and λ_{FC}, respectively) that controls their influence on the final loss. The total loss guides the model to optimize in both CA and VA for accurate predictions, maintaining spatial consistency, and improving the overall quality of the LC map. Mathematically, the HiRes loss is represented by the following equation:

$$\mathcal{L}_{HiRes} = \lambda_{WCE} \times \mathcal{L}_{WCE} + \lambda_{IoU} \times \mathcal{L}_{IoU} + \lambda_{FC} \times \mathcal{L}_{FC}. \tag{9}$$

In this study, we used $\lambda_{WCE} = 1, \lambda_{IoU} = 1$, and $\lambda_{FC} = 0.1$. These values were chosen after preliminary experiments on a held-out subset to balance the influence of pixel-wise accuracy (WCE, IoU) with spatial consistency (FC). The FC loss was given a comparatively low weight because it acts as a regularizer; higher weights led to over-smoothing and boundary blurring of narrow or heterogeneous features (e.g., roads, rivers, and urban fringes), so a small coefficient allows FC to improve consistency without dominating the training signal.

E. Performance Metrics

The performance of the proposed HiResNet model was evaluated using four metrics: precision, recall, $F1$ score ($F1$), and overall accuracy (OA). These metrics were calculated based on the confusion matrix, which consists of four key elements: true positives (TPs), true negatives (TNs), false positives (FPs), and false negatives (FNs). The equations for the metrics are as follows:

$$Precision = \frac{TP}{TP + FP} \tag{10}$$

$$Recall = \frac{TP}{TP + FN} \tag{11}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{12}$$

$$OA = \frac{TP + TN}{TP + TN + FP + FN}. \tag{13}$$

In these equations, TP represents the correctly predicted instances of a specific class. At the same time, TN refers to cases where the model accurately identifies that an example does not belong to a given class. FP occurs when the model incorrectly predicts a class, leading to false alarms. Conversely, FN happens when the model fails to identify an instance that belongs to the target class, resulting in missed detections.

In addition to precision, recall, $F1$, and OA, the receiver operating characteristic (ROC) curve and the area under the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:44:22 UTC from IEEE Xplore. Restrictions apply.

curve (AUC) were employed to further evaluate the HiResNet model's performance in LC classification. The ROC curve is a graphical representation that plots the TP Rate (TPR), or Recall, against the FP Rate (FPR) at varying classification thresholds. The FPR is calculated as follows:

$$FPR = \frac{FP}{TN + FP}. \tag{14}$$

AUC quantifies the HiResNet model's ability to differentiate between classes by measuring the area under the ROC curve. This scalar value serves as a metric for assessing classification performance. An AUC value of one indicates perfect discrimination between classes, while an AUC value of 0.5 suggests no better performance than random guessing. Higher AUC values demonstrate greater model performance in distinguishing between various LC types.

F. Uncertainty Measurement

Understanding and quantifying uncertainty are crucial when generating 10 m LC maps from coarser-resolution (30 m) labels. The process inherently involves challenges, such as resampling of mixed pixels, discrepancies in class assignments, and the influence of spatial and spectral heterogeneity. These factors can lead to uncertainties that impact the accuracy and reliability of the derived 10 m LC map. Uncertainty estimation is particularly important as it helps to identify areas where resampling or upscaling has introduced ambiguity or errors. For example, mixed pixels in the 30 m map, containing contributions from multiple LC types, may not translate precisely into a single class at 10 m, leading to potential misclassifications. Similarly, regions with fragmented landscapes or overlapping spectral characteristics are more prone to classification uncertainty. By measuring uncertainty or visualizing uncertainty maps, we can visualize and assess the spatial distribution of these uncertainties [38]. Accordingly, this study employs class disagreement and entropy-based analysis methods to evaluate the uncertainties in generating the 10 m LC map from the 30 m training labels.

1) Class Disagreement: Class disagreement is a pixel-wise comparison technique used to identify spatial inconsistencies between two maps, typically an HR map and an LR map that has been resampled to match the resolution of HR. This method generates a binary map where each pixel is labeled as either agreement or disagreement based on the correspondence of class values in the two datasets. Specifically, if the class value at a given pixel in the HR map matches the value in the resampled map, it is marked as agreement; otherwise, it is marked as disagreement. Mathematically, the disagreement map is derived using the following equation:

$$D(i,j) = \begin{Bmatrix} 0, & if \quad HR(i,j) = LR(i,j) \\ 1 & if \quad HR(i,j) \neq LR(i,j) \end{Bmatrix} \tag{15}$$

where $HR(i, j)$ represents the class value in the 10 m map at the pixel (i, j) and $LR(i, j)$ represents the class value in the resampled LR map at the same pixel location. To quantify the overall inconsistency between the LR and the generated 10 m maps, the disagreement ratio (dr) is introduced. This ratio measures the proportion of disagreement pixels relative to the total number of pixels (N) in the maps, providing a global metric of spatial inconsistency. It is calculated as follows:

$$dr = \frac{\sum_{i,j} D(i,j)}{N} \tag{16}$$

where $\sum_{i,j} D(i, j)$ represents the total number of pixels with disagreement across the map. The dr provides an overall summary of how well the two maps align, with higher values indicating greater spatial divergence and lower values indicating better agreement. Class disagreement and dr are valuable mechanisms for evaluating the accuracy and reliability of HiResNet model used to generate HR maps from coarser training data. Disagreed areas often indicate regions with mixed pixels, where the LR map contained multiple LC types within a single pixel, leading to potential ambiguities during prediction. In addition, disagreement can arise due to classification inaccuracies in either dataset or discrepancies caused by differences in data collection or processing methods. The visual output of the disagreement map highlights spatial patterns of inconsistency, while the dr value provides a numerical summary of the extent of these inconsistencies.

2) Shannon Entropy: Shannon Entropy is a method used to quantify uncertainty in LC mapping by assessing the unpredictability of class values at each pixel. This study applies entropy to compare the generated HR LC map with the LR LC map. The entropy is computed for each pixel based on the class values from both maps, and it serves as a measure of classification uncertainty, where higher entropy indicates greater ambiguity or mixed class membership. The entropy for each pixel is calculated using the following equation:

$$H = -\sum_{k=1}^{n} p_k \log(p_k + \varepsilon) \tag{17}$$

where H is the entropy value for a given pixel, p_k is the probability of class k in the pixel's value distribution, n is the number of unique classes in the LR pixel, and ε is a small constant added to avoid taking the logarithm of zero. For each pixel (i, j), the class values from HR and LR are treated as the potential outcomes for that pixel. A histogram is computed to estimate the probabilities p_k for each unique class present, and the entropy formula is applied to calculate the level of uncertainty associated with the pixel's classification. The computed entropy values reflect the degree of variability in class assignments between the two maps. In other words, this entropy-based approach provides a statistical assessment of uncertainty across the landscape, identifying areas with high classification ambiguity or inconsistencies between the HR and LR maps.

IV. Results

A. Implementation Detail

The HiResNet model was implemented in TensorFlow (Keras API) within an Anaconda environment. All experiments used 64×64 input patches (15 spectral channels, 11 output classes). We tested batch sizes of 4, 8, 16, and 32 and selected 16 for batch size as it offered the best trade-off between speed and accuracy under our hardware constraints. Training ran

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:44:22 UTC from IEEE Xplore. Restrictions apply.

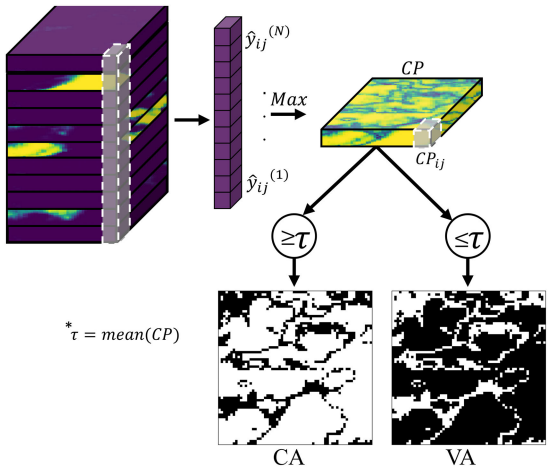


Fig. 9. Diagram illustrates the process of generating CAS module in the HiResNet model. Given the predicted probabilities ($\hat{y}$) from the network, the CA and VA masks are generated based on a dynamic τ which is the mean value of CP. Pixels with probabilities greater than or equal to τ are classified as the CA, while those below τ are classified as the VA.

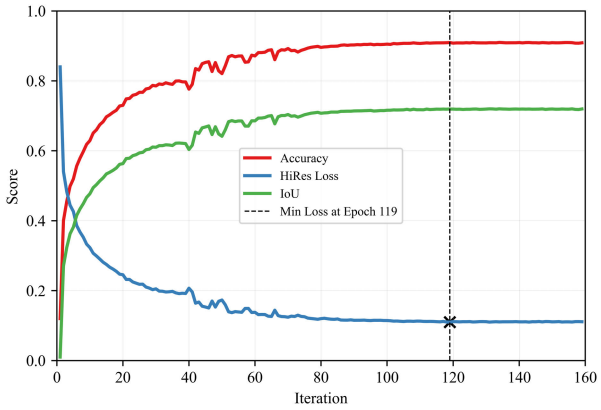


Fig. 10. Training process of the HiResNet model using LR labels.

for up to 1000 epochs with early stopping (patience = 25 epochs, restoring the best weights) and a ReduceLROnPlateau scheduler (factor = 0.5, patience = 5, minimum learning rate of = 10^{-10}) to reduce the learning rate when accuracy plateaued. The network was optimized with Adam (initial learning rate = 5×10^{-4}) and our custom loss function. Model performance was tracked with IoU and OA metrics, and weights of the best-performing weights were saved automatically. During training, on-the-fly data augmentation, consisting of random horizontal/vertical flips and 90° rotations, was applied to increase sample diversity and reduce overfitting. All random seeds were fixed to ensure reproducibility.

B. Training Process

The training of the HiResNet model, as shown in Fig. 10, illustrates the performance trends across 160 iterations. The model's accuracy (red curve) steadily increased and plateaued at approximately 0.91, indicating strong predictive performance. The HiRes loss (blue curve) began at a high value and consistently decreased, reaching its minimum value of 0.11 at epoch 119, as marked by the dashed vertical line. This

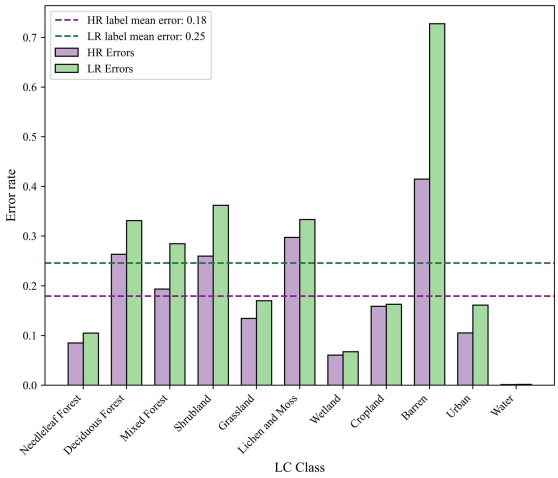


Fig. 11. Comparison of error rates ($1-OA$) between 30 m LR labels and the 10 m LC map generated by HiResNet across different LC classes.

highlights how the model achieved its optimal performance during training. The IoU metric (green curve) also demonstrated continuous improvement, increasing from an initial value of 0.02 to stabilize at around 0.71 toward the end of the training process. These trends reflect the model's effective optimization and enhanced ability to classify and spatially discriminate LC classes accurately.

C. Error Rate

The error rates for the 30 m LR labels and the generated 10 m LC map using the HiResNet model were analyzed, as depicted in Fig. 11. The error rate was calculated as $1 - OA$, where OA represents the accuracy for each LC class. The HR LC map demonstrates a mean error of approximately 0.18, significantly lower than the LR label mean error of around 0.25. As shown, the error rates across different LC classes exhibit noticeable variations, with HR errors consistently lower than LR errors for most classes. Notably, the barren class demonstrates the largest disparity, with HR errors around 0.4 compared to significantly higher errors with LR exceeding 0.7. Among all classes, water bodies have the lowest error rates in both data, with errors approaching 0.01, demonstrating clear spectral separability and highly accurate classification for this class.

D. Quantitative Results

The performance of the HiResNet model was evaluated using Precision, Recall, and $F1$ metrics for each LC class, as summarized in Table II. The OA of the HiResNet model is approximately 0.89, indicating robust performance across various LC classes. Among the classes, water bodies achieve the highest performance metrics with a Precision of 0.96, a Recall of 1.0, and an $F1$ of 0.98. Similarly, wetlands also exhibit strong results, with Precision, Recall, and $F1$ of 0.96, 0.94, and 0.95, respectively, showcasing the model's capability in extracting the boundary of these areas.

Other classes for which the model represents good performance include needleleaf forests, grasslands, and urban

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:44:22 UTC from IEEE Xplore. Restrictions apply.

TABLE II

PERFORMANCE EVALUATION METRICS (PRECISION, RECALL, AND $F1$) FOR EACH LC CLASS OBTAINED FROM THE HIRESNET MODEL USING VALIDATION DATA

Class	Metrices		
	Precision	Recall	F1
Needleleaf forest	0.88	0.91	0.90
Deciduous forest	0.87	0.74	0.80
Mixed forest	0.76	0.81	0.78
Shrubland	0.82	0.74	0.80
Grassland	0.90	0.87	0.88
Lichen and Moss	0.86	0.70	0.78
Wetland	0.96	0.94	0.95
Cropland	0.82	0.84	0.83
Barren	0.96	0.59	0.73
Urban	0.87	0.89	0.88
Water	0.96	1.0	0.98
Total	0.89	0.89	0.89

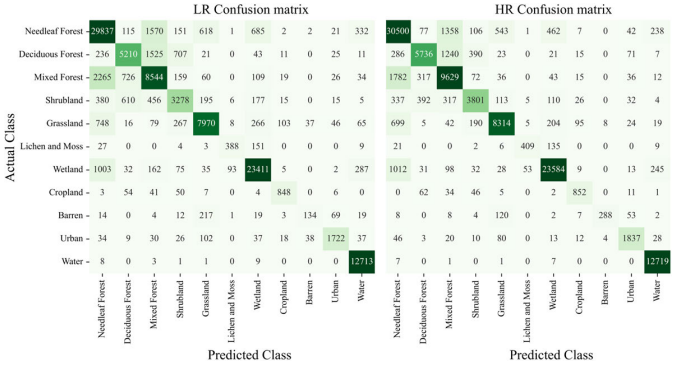


Fig. 12. Comparison of confusion matrices for the LR label and the generated HR LC map using validation data.

areas. These results reflect the model's ability to differentiate these classes accurately, likely due to their unique spectral and structural features. Croplands, shrubland, and deciduous forests also perform well, with an $F1$ of at or exceeding 0.80. However, a slightly lower Recall value of 0.74 for deciduous forests and shrubland suggests the model has minor challenges in classifying the full extent of these classes and misses some instances of these classes.

The model also represents some degree of challenges for classifying the three classes of mixed forest, barren, and lichen and moss. This is potentially due to the spectral similarity of these classes with other existing LC in cases of barren and mixed forest classes, as well as the underrepresentation of lichen and moss and barren classes (see Table I) in our LR dataset.

The confusion matrix is used to understand the performance of the HiResNet model in classifying each class in detail. The confusion matrices for LR labels and HR predictions (see

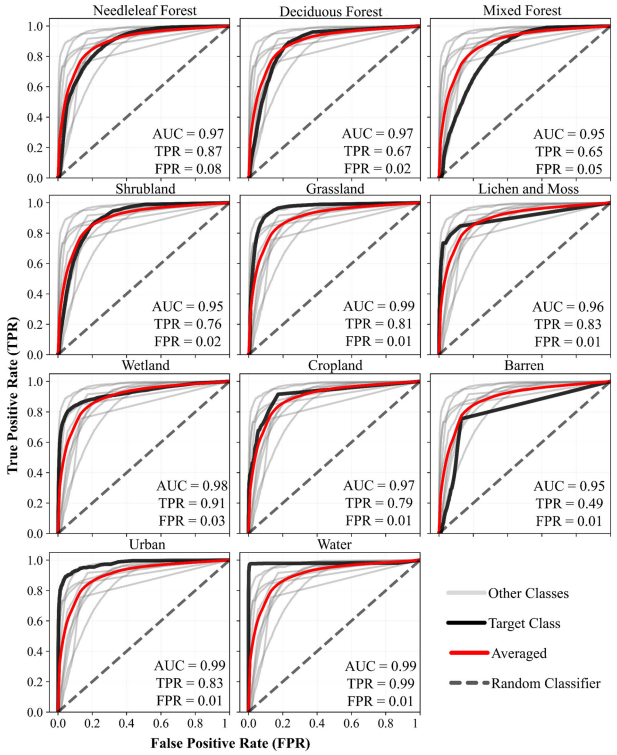


Fig. 13. ROC curves for the HiResNet model across different LC classes.

Fig. 12) illustrate the performance differences between the two LC maps and highlight how the HiResNet model resolves class mixtures in LR data to some extent. In particular, the confusion matrices show the superiority of HR data for classifying all LC classes compared to LR data, as demonstrated in the diagonal elements of the confusion matrices. The improvements are higher for some classes, such as urban, mixed forest, and needleleaf forest, and relatively low for water and cropland classes. The marginal improvement for the later two classes is attributed to the relatively high accuracy of these classes within the LR data.

The ROC curves evaluate the HiResNet model's performance across different LC classes using AUC, TPR, and FPR, as shown in Fig. 13. The results highlight the model's ability to distinguish each target class from others. Notably, the model achieves exceptional performance for water with an AUC of 0.99, a TPR of 0.99, and an extremely low FPR of 0.01, indicating almost perfect classification. Similarly, urban areas exhibit strong results, with an AUC of 0.99, a TPR of 0.83, and a low FPR of 0.01, highlighting the model's ability to classify built-up areas accurately.

Classes like needleleaf forest and wetland also show remarkable results, achieving AUC values of 0.97 and 0.98, respectively. Similarly, for cropland, an AUC of 0.97, a TPR of 0.79, and an FPR of 0.01 demonstrate strong performance in agricultural areas. The model also shows good performance for deciduous forest, lichen and moss, shrubland, and grassland, with AUC values of 0.97, 0.96, 0.95, and 0.99, respectively. The TPRs for these classes range between 0.67 and 0.83, suggesting that while the model effectively identifies these classes, further refinement could enhance its detection capabilities. The

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:44:22 UTC from IEEE Xplore. Restrictions apply.

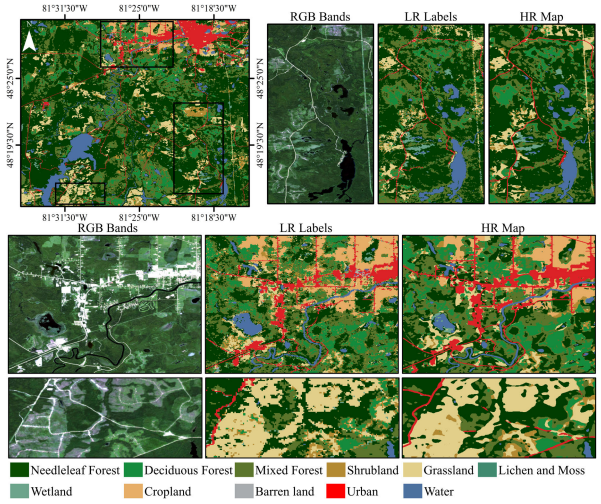


Fig. 14. Qualitative comparison of LR labels and the generated HR LC map using the HiResNet model in forest and urban areas.

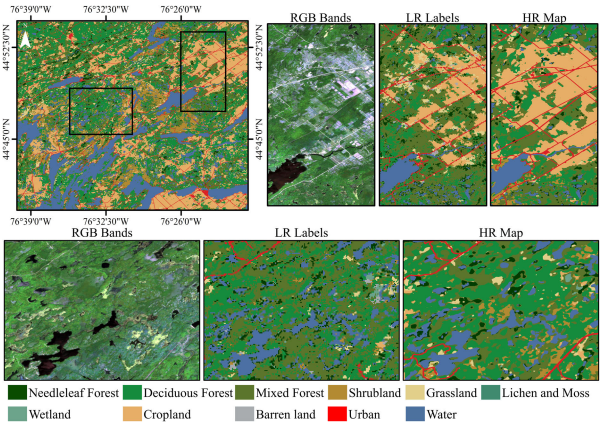


Fig. 15. Qualitative comparison of LR labels and the generated HR LC map using the HiResNet model in cropland, forest, and urban areas.

model, however, represents a moderate performance for barren land with AUC of 0.95, TPR of 0.49, and FPR of 0.01.

E. Qualitative Results

Figs. 14–16 visually compare the output of the proposed method with the LR labels. A true-color composite from Sentinel-2 RGB band is also presented to aid in the interpretation of LR and HR maps. As observed in all figures, the LR labels exhibit noisy classifications due to their coarse pixel size, leading to mismatches in class boundaries, comparatively poor delineation of small-scale features, and imprecise representation of narrow objects, such as rivers and roads. For example, Fig. 15 highlights notable misclassifications and fuzzy boundaries between croplands and forested areas in the LR labels when compared to Sentinel-2 true color composite data.

In contrast, the HR maps generated by the HiResNet model demonstrate superior discrimination capability, with refined class boundaries, improved representation of small-scale features, and better delineation of narrow objects like roads and rivers. Moreover, the HR maps minimize misclassification

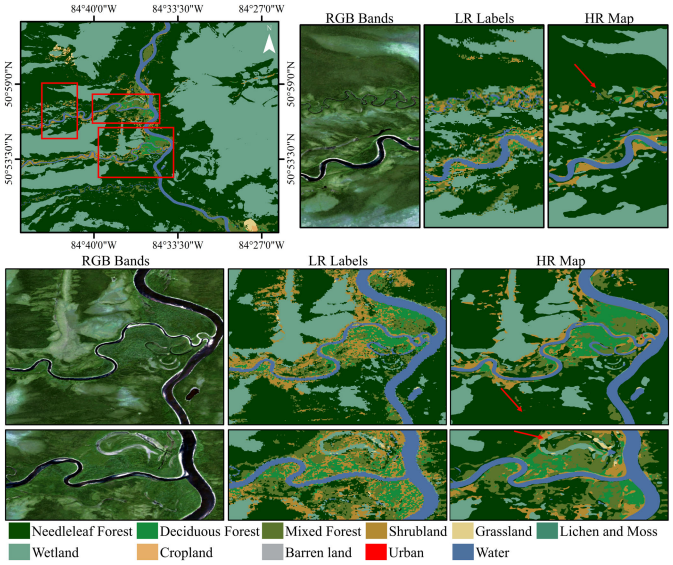


Fig. 16. Qualitative comparison of LR labels and the generated HR LC map using the HiResNet model in wetland, shrubland, and forest regions. The red arrow indicated pixels where the HiResNet model failed to fully extract the feature boundaries.

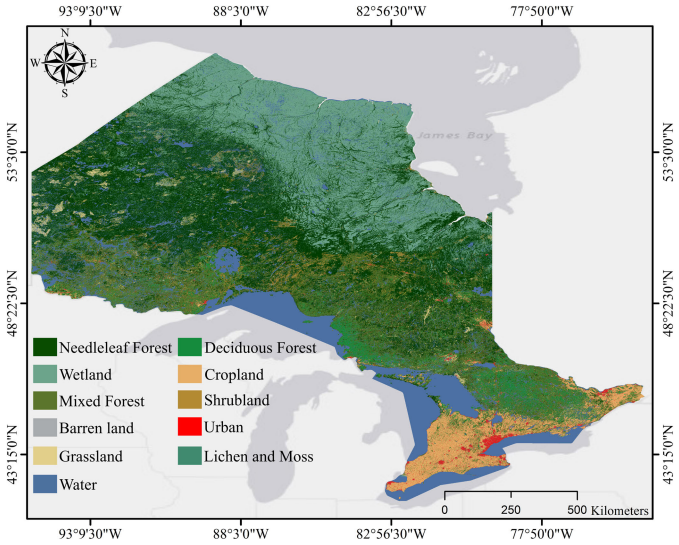


Fig. 17. LC map of ON obtained using the HiResNet model at spatial resolution of 10 m with an OA of 89%.

between vegetated classes (e.g., deciduous forests, shrublands, and grasslands) and extract complex urban structures with a clear-cut boundary. This visual analysis further confirms the capability of the HiResNet model in capturing fine classification details owing to its dual-branch architecture, suitable for both small- and large-scale ground features. Furthermore, with the use of weakly-supervised loss function and application of CA and VA masks, the predictions are less influenced by noisy samples in the coarse labels and more closely align with the true representation of Sentinel-2 imagery. However, despite the detailed landscape classification achieved by the HiResNet model, some misclassified or isolated pixels still remained in HR maps (see Fig. 16, the red arrows). These limitations highlight areas where further refinements in the model or an

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:44:22 UTC from IEEE Xplore. Restrictions apply.

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT MODELS FOR LC MAPPING, EVALUATED USING PRECISION, RECALL, $F1$-SCORE, AND OA

Model	Metric			
	Precision	Recall	F1	OA
RF	0.74	0.81	0.75	0.83
ViT	0.68	0.76	0.80	0.81
HRNet	0.80	0.84	0.82	0.84
DeeplabV3	0.81	0.86	0.83	0.87
SegFormer	0.83	0.85	0.84	0.87
L2HNet	0.86	0.85	0.86	0.87
HiResNet	**0.89**	**0.89**	**0.89**	**0.89**

improvement in the spatial resolution of satellite data (better than 10 m) could enhance the ability to capture complex and narrower features in the landscape. Fig. 17 illustrates the LC map of ON using the HiResNet model with an OA of 89%, featuring 11 LC categories.

F. Model Comparison

In this study, several DL and ML algorithms were used to compare the capability of the HiResNet model with other conventional and advanced techniques for the task of LC mapping. These include random forest (RF) [39], ViT [3], DeepLabV3 [40], SegFormer [41], L2HNet [1], and HRNet [42]. DeepLabV3 used a ResNet-50 [43] backbone with atrous spatial pyramid pooling (ASPP) [35] and DRs of 1, 3, 6, 12, and 18. ViT followed the standard configuration with 16×16 input patches, 768-D embeddings, 12 transformer layers, and 12 self-attention heads. RF was built on per-pixel spectral and index features using 500 trees and unlimited depth. SegFormer, L2HNet, and HRNet were employed using their original architectures as described in their respective papers. For fair comparison, all baseline models were trained on the same data splits and under the same training procedure as HiResNet. Finally, each model was evaluated using the same validation data, enabling a comprehensive analysis of their strengths and weaknesses compared to HiResNet.

Table III compares the performance of baseline models with HiResNet for LC mapping using Precision, Recall, $F1$, and OA. As represented, RF demonstrates a moderate performance, with an $F1$ of 0.75 and an OA of 0.83, given its limited ability to handle the spatial complexity of LC mapping and its reliance on handcrafted features. ViT performs slightly better, in terms of $F1$ (0.80), albeit with a lower OA of 0.81 compared to RF. HRNet demonstrates a good performance, achieving an F1 of 0.82 and an OA of 0.84, as its parallel convolutional branches maintain HR feature representations and enhance spatial precision. DeepLabV3, using atrous convolutions for multiscale feature extraction, achieves improved performance with an $F1$ of 0.83 and an OA of 0.87 by combining local and global contextual features. SegFormer demonstrates comparable capability by achieving an $F1$ of 0.84 and an OA of 0.87 because of its hierarchical transformer encoder and lightweight MLP decoder that efficiently fuses multiscale features while preserving spatial consistency. Likewise, L2HNet attains an $F1$ of 0.86 and an OA of 0.87, demonstrating the effectiveness of its low-to-high learning framework for refining coarse labels into finer spatial resolutions. Despite the promising results obtained by advanced DL models, the proposed model, HiResNet, outperforms other models, achieving the highest $F1$ and OA of 0.89. It confirms its superior capability to integrate multiscale local features and global contextual information through its hybrid CNN–ViT architecture and weakly-supervised learning strategy.

In addition to the quantitative comparison, Fig. 18 compares the classification results of various baseline models with HiResNet in generating HR LC maps from LR labels, along with the true color composite from Sentinel-2 data. As demonstrated, the RF model, while capable of classifying some general patterns, illustrates high levels of misclassification in several cases. This is particularly noticeable for cropland, shrubland, urban, and forest classes, where misclassification occurred.

In many cases, the ViT model performs slightly worse than RF and misclassifies narrow or intricate features such as rivers and small-scale vegetation structures. The SegFormer and DeepLabV3 models show better LC classification capability than RF and ViT. Both models demonstrate superiority in delineating boundaries and extracting small-scale features to reduce noise in the classification outputs. SegFormer produces sharper edges, while DeepLabV3 generates smoother transitions between classes. HRNet maintains consistent spatial precision by preserving HR representations throughout the network, though it tends to produce softer boundaries in heterogeneous regions. L2HNet, which is a noise tolerant model, performs strongly in upscaling coarse labels to finer spatial resolutions. It represents general class distributions effectively but shows limitations in resolving small (roads and rivers) or fragmented features due to its CNN-only architecture. Among all models, HiResNet achieves the most accurate, noiseless, and detailed classification and excels in preserving spatial details and producing clear, well-defined class boundaries.

G. Ablation Study

An ablation study was conducted to evaluate the contribution of different components of the HiResNet model to its overall performance. Ablation studies are systematic evaluation methods used to understand the impact of individual components within a model by selectively removing or modifying them and analyzing the resulting changes in model performance. For HiResNet, the contribution of different components of the proposed model, including the dual DL branches (MSFE and ViT), the proposed loss function, and the CAS threshold, is evaluated. To quantify the effects of these components, various scenarios are examined and represented in Table IV.

As represented in Table IV, the original HiResNet model, with its full architecture and the HiRes loss function, achieved

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:44:22 UTC from IEEE Xplore. Restrictions apply.

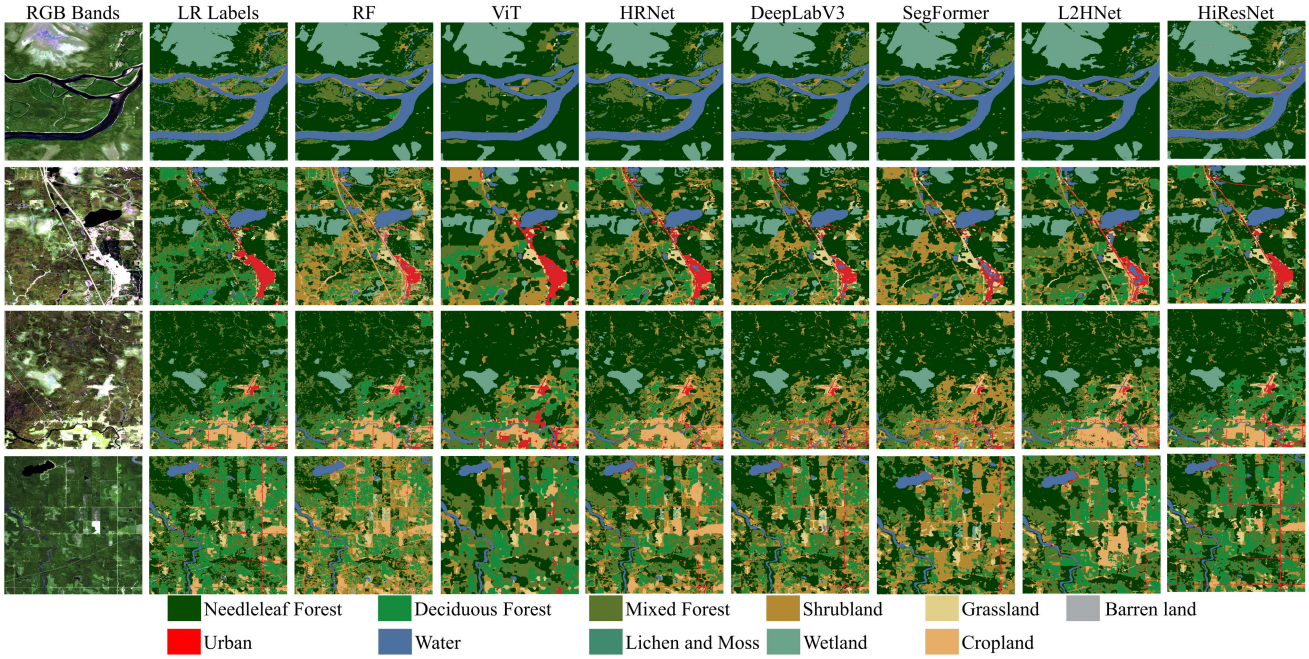


Fig. 18. Visual comparison of LC classification results generated by different models (RF, ViT, HRNet, DeepLabV3, SegFormer, L2HNet, and HiResNet) alongside LR labels and Sentinel-2 RGB bands.

the highest $F1$ of 0.89, serving as the benchmark for comparison. Removing the MSFE component (HiResNet^1) led to a significant performance drop, with an $F1$ of 0.80, underscoring the importance of MSFE in extracting multiscale spatial features. Similarly, excluding the ViT module (HiResNet^2) resulted in an $F1$ of 0.82, indicating that the ViT component plays a crucial role by capturing long-range contextual dependencies within the input data. Replacing the MSFE with a simple CNN (HiResNet^3) achieved an $F1$ of 0.81, comparable to HiResNet^2, yet still remained below that of the original HiResNet.

Modifications to the loss function also significantly impacted the $F1$ metrices. Training the model using only the IoU loss ($\mathcal{L}_{HiRes}^{IoU}$) resulted in an $F1$ of 0.76 while using only the WCE loss ($\mathcal{L}_{HiRes}^{WCE}$) resulted in a better $F1$ of 0.82. However, relying solely on the unsupervised FC loss ($\mathcal{L}_{HiRes}^{FC}$) significantly affected the accuracy, with an $F1$ of just 0.33, demonstrating its inefficiency for training if used by itself. Among the pairwise combinations, the best performance was observed with IoU and WCE losses ($\mathcal{L}_{HiRes}^{IoU+WCE}$) achieving an $F1$ of 0.85, which outperformed IoU and FC ($\mathcal{L}_{HiRes}^{IoU+FC}$) and WCE and FC ($\mathcal{L}_{HiRes}^{FC+WCE}$), with $F1$ of 0.79 and 0.80, respectively. The full HiRes loss function ($\mathcal{L}_{HiRes}^{FC+CE+IoU}$) excluding the weighting mechanism (Scenario 10), resulted in an $F1$ of 0.86. However, the proposed HiRes loss function with its weighting mechanism with its full architecture achieved the highest $F1$ of 0.89. This result further confirms that the designed loss function, which balances contributions from IoU, FC, and WCE, significantly improves the model's ability to handle the complexities of LC mapping.

In addition, an ablation analysis was conducted to examine the impact of the proposed CAS dynamic thresholding strategy. Three fixed confidence thresholds at 0.25, 0.50, and 0.75 were evaluated and compared against the proposed dynamic CAS formulation. As shown in Table IV, fixed thresholds result in stable but consistently lower performance compared to the dynamic strategy. Lower fixed thresholds allow a larger number of low-confidence pixels, thereby increasing sensitivity to noisy supervision, whereas higher fixed thresholds overly restrict the supervision signal and limit the effective use of available training data. Although increasing the fixed threshold improves the $F1$ score from 0.81 to 0.85, none of the fixed configurations achieve the performance of the dynamic CAS strategy, which achieves the highest $F1$ score of 0.89. These results confirm that the proposed dynamic thresholding strategy provides a more effective balance between robustness to label noise and supervision coverage, leading to more stable optimization and improved overall performance of the model.

V. Discussion

A. Uncertainty Estimation Result

In LC mapping using remote sensing data, uncertainty arises from various sources, including input data, model architecture, ground truth labels, and even validation approaches [44]. Among these factors, the quality of ground truth data used during the training phase plays a pivotal role. Low-quality label data results in misclassifications and deviations in the model's predictions. In this study, where LR training labels were used to produce an HR LC map, quantifying the degree of uncertainty becomes crucial for better understanding the model's reliability and addressing potential errors stemming from imperfect labels.

Disagreement between the generated 10 m LC map and the resampled 30 m labels is largely driven by resolution-induced effects, including mixed pixels, sub-pixel heterogeneity, and boundary aggregation inherent to coarse-resolution labels. Consequently, LR–HR disagreement does not necessarily indi-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:44:22 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
RESULTS OF THE ABLATION STUDY ON THE HIRESNET MODEL, COMPARING DIFFERENT ARCHITECTURAL COMPONENTS, LOSS FUNCTION CONFIGURATIONS, AND THE CAS THRESHOLD STRATEGIES

Variant	Scenario	Description	Abbreviation	F1
HiResNet architecture	1	Exclude the MSFE from HiResNet architecture	HiResNet^{1}	0.80
	2	Exclude the ViT module from HiResNet architecture	HiResNet^{2}	0.82
	3	Replace the MSFE with a simple CNN	HiResNet^{3}	0.81
Loss function	4	Training the HiResNet model using only the IoU loss	$\mathcal{L}_{HiRes}^{IoU}$	0.76
	5	Training the HiResNet model using only the FC loss	$\mathcal{L}_{HiRes}^{FC}$	0.33
	6	Training the HiResNet model using only the WCE loss	$\mathcal{L}_{HiRes}^{WCE}$	0.82
	7	Training the HiResNet model with IoU and FC losses	$\mathcal{L}_{HiRes}^{IoU+FC}$	0.79
	8	Training the HiResNet model with IoU and WCE losses	$\mathcal{L}_{HiRes}^{IoU+WCE}$	0.85
	9	Training the HiResNet model with WCE and FC losses	$\mathcal{L}_{HiRes}^{FC+WCE}$	0.80
	10	Training the HiResNet model with IoU, FC, and CE losses	$\mathcal{L}_{HiRes}^{IoU+FC+CE}$	0.86
CAS threshold strategy	11	Fixed CAS threshold = 0.25	HiResNet$^{0.25}$	0.81
	12	Fixed CAS threshold = 0.5	HiResNet$^{0.5}$	0.83
	13	Fixed CAS threshold = 0.75	HiResNet$^{0.75}$	0.85
The original proposed model	14	The original HiResNet model with HiRes loss function	HiResNet	0.89

cate incorrect model predictions, but rather highlights spatial inconsistencies introduced during the resolution transition from 30 to 10 m. This type of inconsistency is expected in heterogeneous landscapes and along narrow or fragmented features that cannot be fully represented at 30 m resolution. In contrast, model classification uncertainty is associated with ambiguity in the predicted class probabilities, which arises from spectral overlap, within-class variability, and noisy supervision. In this study, entropy-based uncertainty is used to represent model-related uncertainty, while class disagreement

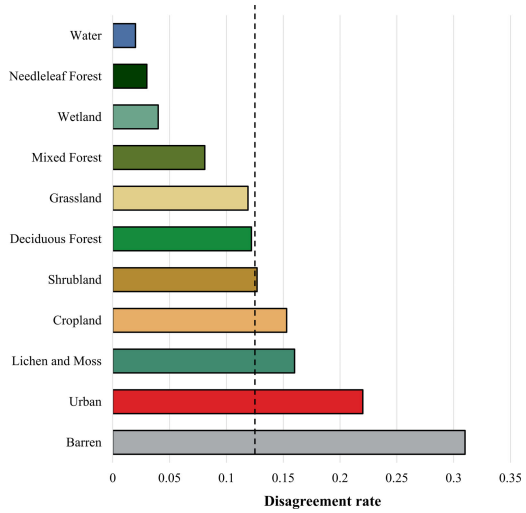


Fig. 19. Class disagreement rate for each individual LC class.

serves as a complementary indicator of inconsistency induced by resolution. Interpreting these measures provides a more complete understanding of uncertainty in a weakly-supervised HR LC mapping.

1) Spatial Distribution of Disagreement: The spatial distribution of disagreement shows which regions and LC classes do not match within HR and LR LC maps. This measure represents the ratio of matching pixel values between the LR labels and the generated HR LC map, providing a quantitative assessment of the model's confidence. Overall, in remote sensing LC applications, the edges of objects tend to have the highest levels of uncertainty [38], [45]. This issue is particularly pronounced when distinguishing between spectrally similar classes, such as wetlands, shrublands, and various types of forest classes.

The class disagreement approach was applied across the study area where ground truth samples are located and have the same classes as the generated HR map to evaluate disagreement for each LC class individually, as depicted in Fig. 19. The overall disagreement level across all classes was 0.125, indicating that approximately 12% of pixels in the LR and generated HR LC maps do not match. This level of disagreement can be attributed to the complex landscape, the presence of small or narrow objects, and challenges associated with the LR labels. For instance, small wetlands and shrubland objects appear in the generated HR LC map but are absent in the LR map. Similarly, narrow features such as roads and rivers are frequently omitted in the LR LC map, primarily due to their widths being less than 30 m, which falls below the resolution threshold of the LR data.

Despite these challenges, the HiResNet model, equipped with the MSFE module, effectively demonstrates its capacity to classify objects of varying scales. The results confirm the models' ability to identify small and narrow objects otherwise overlooked in the LR data. As illustrated in Fig. 19, the two classes of barren and urban areas exhibited the highest levels of disagreement. This can be partially explained by the spectral and structural similarities between barren land and urban regions, leading to confusion in classification. In the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:44:22 UTC from IEEE Xplore. Restrictions apply.

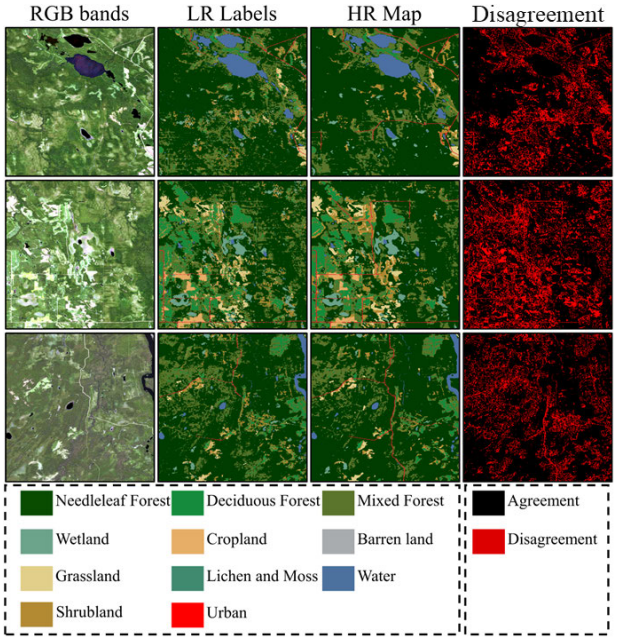


Fig. 20. Three visual examples of class disagreement: The first column illustrates RGB bands derived from Sentinel-2 data, the second column demonstrates LR labels, the third column shows generated HR maps, and the last column depicts disagreement maps.

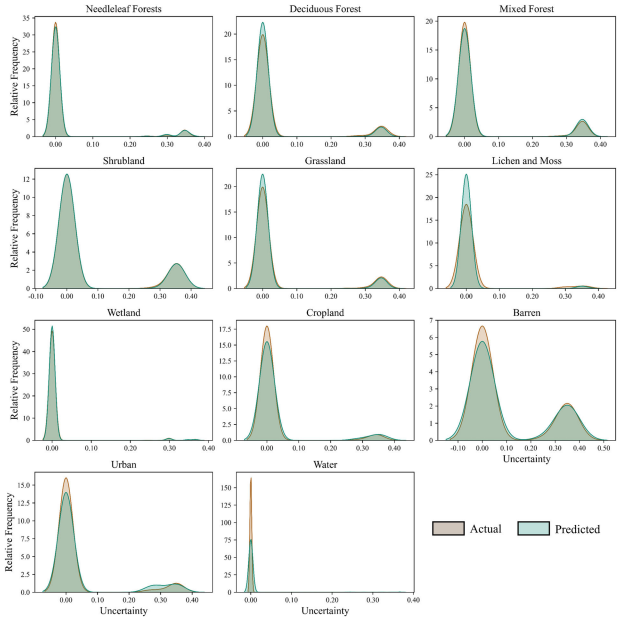


Fig. 21. Distribution of the relative frequency of Shannon entropy values for uncertainty analysis across different LC classes for actual (orange color) and predicted (green color) values.

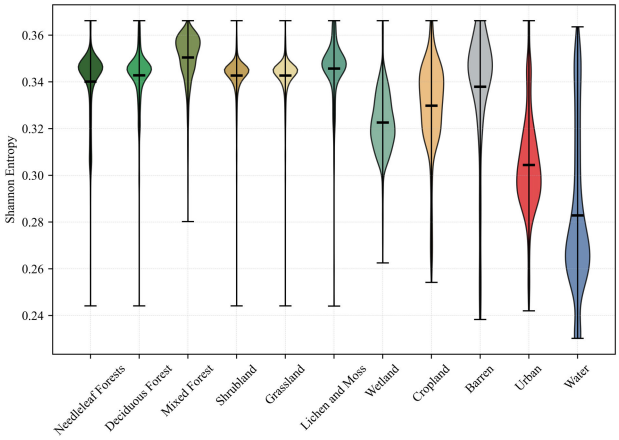


Fig. 22. Distribution of the Shannon entropy values across the whole region (provincial boundary) between the resulting 10 m HR and the existing 30 m LR maps.

case of the urban class, the visual assessment of the produced map confirmed better discrimination of this class using the HR data owing to its higher spatial resolution, which better matches with the characteristics of the urban landscape. In particular, the 30 m resolution of the LR LC map resulted in the omission of fine details, such as roads and small buildings, contributing significantly to the high disagreement observed in these classes.

Moreover, lichen and moss, croplands, shrublands, deciduous forests, and grasslands exhibited class disagreement rates ranging between 0.16 and 0.12. This disagreement can be attributed to overlapping spectral characteristics, subtle differences between these LC types, and noisy pixels in the LR map (as shown in Fig. 20). The results show wetlands, despite their known complexity, exhibited a higher degree of agreement between the LR and generated HR LC maps, confirming a relatively good classification of this class within both data that agree with the validation data. Furthermore, the two classes of needleleaf forests and water exhibited the lowest levels of disagreement among all LC classes. This could be also due to the well representation of these two classes in the LR LC training data, as well as the relatively clear spectral signature of these classes compared to other existing LC types in the study region. It is also expected that the derived disagreement in the water class primarily arises from narrow rivers and small lakes, which are challenging to capture in the LR data at 30 m spatial resolution.

Three examples of class disagreement are illustrated in Fig. 20, providing visual confirmation of our findings. These examples highlight the discrepancies between the LR and generated HR LC maps, showcasing areas where the model addressed existing issues, such as mixed pixels, overlapping spectral characteristics, or the limitations of coarse-resolution data.

2) Quantifying Uncertainty in Class Mixtures: When using 30 m labels to train models on 10 m data, a single 30 m pixel corresponds to a maximum of nine 10 m pixels. The diversity of class labels within 30 m pixels directly contributes to the uncertainty when the LR labels are used for HR LC classification. If all nine 10 m pixels belong to different classes, this pixel represents the highest possible uncertainty, indicating significant heterogeneity. Conversely, if all nine 10 m pixels share the same class, the uncertainty is zero, suggesting complete homogeneity. This phenomenon highlights a key limitation of 30 m labels, where multiple objects within a single pixel are aggregated into a single class.

This uncertainty is critical in scenarios where the HiResNet model identifies objects within the 10 m resolution that are not present in the coarser 30 m labels. Point validation data was used to quantify this level of uncertainty, and Shannon entropy

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:44:22 UTC from IEEE Xplore. Restrictions apply.

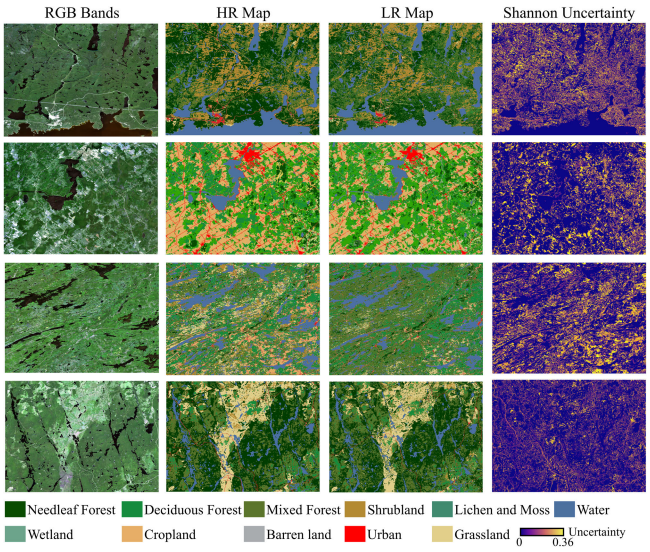


Fig. 23. Four visual examples of Shannon entropy uncertainty. The first column demonstrates RGB images (Sentinel-2), the second column demonstrates generated HR LC map, the third column demonstrates LR labels, and the last column demonstrates Shannon entropy uncertainty.

values were computed for the points where validation data existed. Shannon entropy measures the diversity or mixture of classes within a pixel, considering the inherent ambiguity in the 30 m label's representation. We analyzed the entropy distributions for actual and predicted maps across various LC classes (see Figs. 21–23).

To interpret and discuss the findings more consistently, LC classes were categorized into three uncertainty levels: very low, low, and moderate. This grouping enables a clearer understanding of how class properties affect entropy. Very low uncertainty was observed for urban, water, and wetland classes. These classes consistently show the lowest entropy values across all three figures (Figs. 21–23). Urban and water areas are generally spectrally distinct and spatially cohesive, making them easier to classify with minimal class mixture. The observed entropy in urban and water classes is mostly due to narrow or small-scale features such as roads, small buildings, and rivers that are not fully extracted in 30 m labels. Wetlands, despite being ecologically complex, show low uncertainty/entropy in our results, likely due to the ability of the model to discriminate their boundaries and to accurately identify their spectral characteristics with HR data.

Low uncertainty was found in needleleaf forest, deciduous forest, shrubland, grassland, and cropland. Forest classes, in particular, exhibit strong agreement between actual and predicted entropy values, as shown in Fig. 21, with distinct peaks aligned across both maps. This can be attributed to their spatially continuous nature. While Fig. 22 indicates some variation in entropy for these classes due to interal heterogeneity and edge effects, overall uncertainty remains low. Shrubland, grassland, and cropland display slightly elevated entropy compared to forests, particularly in fragmented landscapes or where mixed vegetation types are present within the same 30 m pixel. Nonetheless, the model handles these classes effectively, and entropy distributions between actual and predicted maps remain aligned.

Moderate entropy is primarily observed for lichen and moss, barren land, and mixed forest. The lichen and moss, and barren classes often occur in fragmented patches or transitional zones and lack strong, consistent spectral signatures. As a result, they exhibit broader entropy distributions in both the actual and predicted maps. In the case of mixed forest, the observed uncertainty may arise from spectral similarities with needleleaf forest, as also indicated in the confusion matrix (Fig. 12). Fig. 23 illustrates these variations through representative examples, highlighting the model's challenges in consistently resolving these classes at finer spatial scales. Despite these difficulties, the entropy patterns remain reasonable and demonstrate the model's overall robustness.

To support the actionable use of uncertainty, we further examined where high-entropy and high-disagreement pixels occur and how they correlate with classification errors. We found that >80% of high-entropy pixels lie along class boundaries, as well as in spectrally mixed areas (urban–vegetation edges, small water bodies), and in regions prone to misclassification. For end-users who wish to update land cover inventories, the uncertainty maps can serve as a quality-control layer. For example, setting an entropy threshold of 0.8 flags ~5% of pixels for manual review (visual interpretation of HR imagery), targeted field surveys, or use them as a priority pool for active learning, where new labels are collected preferentially in uncertain areas to improve model performance iteratively. Providing these uncertainty layers alongside the land cover maps offers a practical quality-control and label-refinement tool without requiring additional computational effort.

B. Model Generalization and Scalability

The generalization capability of a DL model is a critical metric that reflects its ability to perform well on unseen datasets beyond the training region. In the context of large-scale LC mapping, achieving generalization is particularly challenging due to the inherent diversity of landscapes and scenes. Errors and misclassifications can arise when models encounter LC types, spatial patterns, or spectral features not adequately represented in the training dataset. These biases can significantly reduce the model's performance across large and heterogeneous regions.

Previous studies [46] have demonstrated the potential of supervised algorithms for large-scale LC mapping. However, they primarily relied on high-quality training labels, which are expensive and time-intensive to collect, particularly over expansive areas. Our study addresses this limitation by reducing the dependence on high-quality labels. We introduce a novel DL architecture, HiResNet, coupled with an innovative weakly-supervised loss function that uses LR labels. Using LR training labels, our approach offers a cost-effective solution without compromising the accuracy and scalability of LC mapping.

In contrast to conventional ML algorithms, such as RF and support vector machines (SVMs), frequently implemented

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:44:22 UTC from IEEE Xplore. Restrictions apply.

in cloud platforms like GEE [47], and extensively used for large-scale studies, our methodology adopts the state-of-the-art DL framework. While platforms like GEE provide robust computational resources for ML models, their constraints often make deploying DL algorithms infeasible. Our study overcomes this limitation by successfully applying a large-scale DL model for LC mapping, offering a more robust and accurate solution to large-scale challenges.

Li et al. [1] also developed and applied the L2HNet framework based on a CNN architecture for large-scale LC mapping. While their work demonstrated good results in enhancing spatial resolution and classification accuracy, the scale of their study area was significantly smaller, covering approximately 33 872 km^2. This scale is not directly comparable to the scope of our study, which spans approximately 1.07 million km^2. The vast differences in the size of study areas introduce additional challenges in our research, including a higher landscape diversity, greater variability in LC types, and increased computational demands. Large-scale LC mapping over such extensive areas requires robust model architecture and innovative methodologies to handle the heterogeneity and complexity of the landscape.

While HiResNet demonstrates strong scalability in terms of spatial generalization, practical deployment at the provincial scale also involves nontrivial computational considerations. In this study, the total volume of processed Sentinel-2 imagery and intermediate products exceeded 400 GB, reflecting the high spatial resolution (10 m) and large geographic extent (~1.07 million km^2). Model training required approximately 4–6 days, depending on architectural configuration and loss-function settings, while full-province inference required approximately 15 days. These runtimes primarily reflect the scale of the data and the weakly-supervised learning paradigm rather than inefficient model design.

HiResNet is intended for offline or periodic land cover inventory updates, rather than real-time inference. The adopted tile-based processing strategy allows inference to be trivially parallelized across computing nodes, making the framework practical for large-area deployment when sufficient resources are available. In this context, the selection of tile size is a critical deployment parameter, as it directly affects memory usage, processing time, and scheduling efficiency. Larger tiles provide broader spatial context but increase memory demand, whereas smaller tiles reduce memory requirements at the cost of increased boundary handling and orchestration overhead. The tile dimensions used in this study represent a practical balance, considering the available computational resources.

Compared to lightweight architectures such as SegFormer, HiResNet incurs higher computational cost due to its dual-branch design and modeling of multiscale context. However, this additional cost enables improved robustness and accuracy under coarse and noisy supervision, particularly in heterogeneous landscapes. The choice between HiResNet and lighter baselines therefore reflects a trade-off between computational efficiency and mapping fidelity, depending on application requirements.

C. Principal Drivers of HiResNet's Performance

The superior performance of HiResNet for LC mapping can be attributed to several principal drivers integral to its architecture and training methodology. A key training component is the weakly-supervised HiRes loss, specifically designed for such a heterogeneous landscape, by combining WCE, IoU, and feature-consistency (FC) terms under a confidence-area selection mask. This custom loss function proved to be excelled in handling class imbalances and complex boundary ambiguities. In particular, unlike conventional losses that treat every pixel equally, this formulation reduces weights on ambiguous or misaligned pixels from the 30 m labels while still exploiting their contextual information. This mechanism enables the model to extract sharper boundaries and fine structures from LR supervision and thus reduces its reliance on HR training labels, which has been confirmed with the ablation results presented in Table IV.

The use of Sentinel-2 imagery with its 10 m spatial resolution often leads to mixed spectral responses when aligned with 30 m LC labels; the MSFE module addresses this issue by learning features at multiple scales. This capability enables the model to differentiate between LC classes accurately despite the spectral and spatial overlaps inherent to heterogeneous landscapes and to solve boundary ambiguities, particularly in complex regions such as urban–vegetation interfaces or fragmented landscapes, thereby enhancing accuracy and robustness.

Similar studies [1], [19] also used multikernel CNN blocks to consider the differences between HR input data and 30 m labels. In LC remote sensing tasks, the MSFE module with varying DRs is superior to a multikernel approach in CNNs [34], [36], [43]. This is because DRs enable the network to accurately capture spatial features across multiple scales without significantly increasing the number of parameters or computational overhead. By varying DRs, the receptive field of convolutional filters expands, allowing the network to integrate context from both fine and coarse spatial resolutions in a single layer. In contrast, a multikernel approach requires separate convolutional filters for each kernel size, leading to increased model complexity and parameter count. In addition, the MSFE module with varying DRs avoid the redundancy of overlapping filters seen in multikernel approaches, making the model to be more efficient in extracting diverse spatial features. These properties make dilation-based feature extraction particularly well-suited to the hierarchical and multiscale nature of heterogenous LC mapping tasks using satellite imagery, especially when using LR labels.

Another key driver of HiResNet's performance is its hybrid architecture, which combines MSFE and ViT. This combination allows the model to integrate fine spatial details and global contextual information simultaneously. Removing the ViT module in the ablation experiment, conducted in this research, resulted in a lower performance, demonstrating its contribution in maintaining a global perspective while ensuring accurate classification of large-scale LC maps (see Table IV).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:44:22 UTC from IEEE Xplore. Restrictions apply.

D. Limitations

The HiResNet model demonstrates strong performance in large-scale, detailed LC mapping; however, several limitations remain that could be addressed to improve its efficiency and robustness. One challenge is the computational demand associated with its high-capacity modules, particularly the MSFE and ViT components, which, while effective in enhancing feature representation, increase the model's complexity. To mitigate this, future work could explore more lightweight alternatives, such as MobileViT-like architectures [28], which preserve the benefits of both convolutional and transformer-based designs with fewer parameters, making the model more suitable for near real-time or resource-constrained applications. In addition, issues related to class imbalance still may persist despite employing a composite loss function that integrates WCE, IoU, and FC losses. Rare and underrepresented classes, such as lichen and moss and barren, or narrow rivers, remain more prone to misclassification due to limited representation in the training data. Addressing this may require further strategies like focal loss tuning, adaptive sampling, or uncertainty-aware training.

Another limitation arises from the inherent ambiguity in training data. HiResNet relies on 30 m LC labels for training, which may aggregate multiple LC types into a single pixel, introducing uncertainty, particularly in heterogeneous regions. While the MSFE module helps to mitigate this issue by learning features at multiple scales, the reliance on coarse-resolution labels restricts the model's ability to fully leverage the 10 m resolution of Sentinel-2 imagery, particularly for extracting fine details, such as narrow roads, small buildings, or fragmented vegetation patches.

The generalization capability of HiResNet across diverse landscapes or climatic zones is another area of concern. While the model has performed well within the study area, its effectiveness in regions with vastly different spectral and spatial characteristics remains untested. Fine-tuning the model with additional region-specific data could address this issue and enhance its adaptability. Furthermore, the complexity of the weakly-supervised loss function, with its reliance on multiple hyperparameters, may pose challenges for fine-tuning and adapting the model to new datasets or applications.

VI. Conclusion

This study introduced HiResNet, an innovative deep-learning framework, designed for generating HR (10 m) LC maps from an existing LR (30 m) LC map, as training labels, using Sentinel-2 satellite data. HiResNet has a hybrid architecture, leveraging CNN and ViT branches for complex, fragmented LC mapping. The CNN branch consists of an MSFE module, well-suited for extracting multiscale spatial features from a heterogeneous landscape. On the other hand, the ViT branch extracts global features within input images. However, the core of the HiReNet model is its weakly-supervised loss function that overcomes the challenges of HR image segmentation, uncertainty in predictions, and potential class imbalances.

The HiResNet model demonstrated superior performance in large-scale LC mapping, achieving an OA of 0.89 across a diverse and extensive landscape in ON, Canada. The results highlighted the HiResNet's robustness in accurately classifying complex LC types, such as wetlands and mixed forests, and resolving boundary ambiguities for narrow or fragmented features like roads and rivers. In addition, the study explored uncertainty estimation techniques, revealing critical understandings of areas of high disagreement between LR 30 m labels and generated HR 10 m maps, often due to the inherent limitations within coarse training data.

The proposed framework offers a scalable and efficient solution for updating large-scale LC inventories at finer spatial resolutions without requiring extensive, high-quality training data. However, limitations related to narrow feature extraction and class separability in heterogeneous regions indicate areas for further refinement. Future studies could explore fine-tuning the model on region-specific data, incorporating additional spectral indices, or using advanced explainable AI techniques to enhance model transparency and performance.

References

[1] Z. Li, H. Zhang, F. Lu, R. Xue, G. Yang, and L. Zhang, "Breaking the resolution barrier: A low-to-high network for large-scale high-resolution land-cover mapping using low-resolution labels," *ISPRS J. Photogramm. Remote Sens.*, vol. 192, pp. 244–267, Oct. 2022, doi: 10.1016/j.isprsjprs.2022.08.008.

[2] C. Gómez, J. C. White, and M. A. Wulder, "Optical remotely sensed time series data for land cover classification: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 116, pp. 55–72, Jun. 2016, doi: 10.1016/j.isprsjprs.2016.03.008.

[3] M. Marjani, F. Mohammadimanesh, M. Mahdianpari, and E. W. Gill, "A novel spatio-temporal vision transformer model for improving wetland mapping using multi-seasonal sentinel data," *Remote Sens. Appl., Soc. Environ.*, vol. 37, Jan. 2025, Art. no. 101401.

[4] M. A. M. Choudhury, E. Marcheggiani, A. Galli, G. Modica, and B. Somers, "Mapping the urban atmospheric carbon stock by LiDAR and WorldView-3 data," *Forests*, vol. 12, no. 6, p. 692, May 2021, doi: 10.3390/f12060692.

[5] M. Marjani, M. Mahdianpari, F. Mohammadimanesh, and E. W. Gill, "CVTNet: A fusion of convolutional neural networks and vision transformer for wetland mapping using Sentinel-1 and Sentinel-2 Satellite data," *Remote Sens.*, vol. 16, no. 13, p. 2427, Jul. 2024.

[6] F. Mohammadimanesh, B. Salehi, M. Mahdianpari, E. Gill, and M. Molinier, "A new fully convolutional neural network for semantic segmentation of polarimetric SAR imagery in complex land cover ecosystem," *ISPRS J. Photogramm. Remote Sens.*, vol. 151, pp. 223–236, May 2019, doi: 10.1016/j.isprsjprs.2019.03.015.

[7] A. Lönnqvist, Y. Rauste, M. Molinier, and T. Häme, "Polarimetric SAR data in land cover mapping in boreal zone," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 10, pp. 3652–3662, Oct. 2010, doi: 10.1109/TGRS.2010.2048115.

[8] G. De Luca, J. M. N. Silva, S. Di Fazio, and G. Modica, "Integrated use of Sentinel-1 and Sentinel-2 data and open-source machine learning algorithms for land cover mapping in a Mediterranean region," *Eur. J. Remote Sens.*, vol. 55, no. 1, pp. 52–70, Dec. 2022, doi: 10.1080/22797254.2021.2018667.

[9] Y. Chen, X. He, J. Wang, and R. Xiao, "The influence of polarimetric parameters and an object-based approach on land cover classification in coastal wetlands," *Remote Sens.*, vol. 6, no. 12, pp. 12575–12592, Dec. 2014, doi: 10.3390/rs61212575.

[10] M. Mahdianpari, B. Salehi, F. Mohammadimanesh, and M. Motagh, "Random forest wetland classification using ALOS-2 L-band, RADARSAT-2 C-band, and TerraSAR-X imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 130, pp. 13–31, Aug. 2017, doi: 10.1016/j.isprsjprs.2017.05.010.

[11] A. A. Saim and M. Aly, "Enhancing tree species mapping in Arkansas' forests through machine learning and satellite data fusion: A Google Earth engine-based approach," *J. Geovisualization Spatial Anal.*, vol. 9, no. 1, p. 20, Jun. 2025, doi: 10.1007/s41651-025-00220-9.

[12] G. Amin, I. Imtiaz, E. Haroon, N. U. Saqib, M. I. Shahzad, and M. Nazeer, "Assessment of machine learning algorithms for land cover classification in a complex mountainous landscape," *J. Geovisualization Spatial Anal.*, vol. 8, no. 2, p. 34, Dec. 2024, doi: 10.1007/s41651-024-00195-z.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:44:22 UTC from IEEE Xplore. Restrictions apply.

[13] P. Dou, H. Shen, C. Huang, Z. Li, Y. Mao, and X. Li, "Large-scale land use/land cover extraction from Landsat imagery using feature relationships matrix based deep-shallow learning," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 129, May 2024, Art. no. 103866, doi: 10.1016/j.jag.2024.103866.

[14] P. Dou, C. Huang, W. Han, J. Hou, Y. Zhang, and J. Gu, "Remote sensing image classification using an ensemble framework without multiple classifiers," *ISPRS J. Photogramm. Remote Sens.*, vol. 208, pp. 190–209, Feb. 2024.

[15] R. D. Morton, C. G. Marston, A. W. O'Neil, and C. S. Rowland, "Land Cover Map 2017 (25m rasterised land parcels, GB)," NERC Environ. Inf. Data Centre, U.K. Centre Ecol. Hydrol., Tech. Rep., 2020. [Online]. Available: https://doi.org/10.5285/499212cd-d64a-43ba-b801-95402e4d4098

[16] J. Wickham, S. V. Stehman, D. G. Sorenson, L. Gass, and J. A. Dewitz, "Thematic accuracy assessment of the NLCD 2016 land cover for the conterminous United States," *Remote Sens. Environ.*, vol. 257, May 2021, Art. no. 112357.

[17] M. Pasos. *NALCMS—The North American Land Change Monitoring System*. Accessed: Oct. 29, 2025. [Online]. Available: https://www.cec.org/publications/nalcms/

[18] J. Chen et al., "Global land cover mapping at 30 m resolution: A POK-based operational approach," *ISPRS J. Photogramm. Remote Sens.*, vol. 103, pp. 7–27, May 2015.

[19] Z. Li, W. He, J. Li, F. Lu, and H. Zhang, "Learning without exact guidance: Updating large-scale high-resolution land cover maps from low-resolution historical labels," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, May 2024, pp. 27717–27727. [Online]. Available: http://openaccess.thecvf.com/content/CVPR2024/html/Li_Learning_without_Exact_Guidance_Updating_Large-scale_High-resolution_Land_Cover_Maps_CVPR_2024_paper.html

[20] M. Marjani, M. Mahdianpari, E. W. Gill, and F. Mohammadimanesh, "A dual-branch deep learning framework for wetland mapping using Sentinel-1 and Sentinel-2," in *Proc. IEEE 20th Int. Symp. Antenna Technol. Appl. Electromagn. (ANTEM)*, Jul. 2025, pp. 246–249. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/11114179/

[21] N. Alizadeh, M. Mahdianpari, E. Hemmati, and M. Marjani, "FusionFireNet: A CNN-LSTM model for short-term wildfire hotspot prediction utilizing spatio-temporal datasets," *Remote Sens. Appl., Soc. Environ.*, vol. 37, Jan. 2025, Art. no. 101436.

[22] C. Robinson et al., "Large scale high-resolution land cover mapping with multi-resolution data," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 12718–12727. [Online]. Available: http://openaccess.thecvf.com/content_CVPR_2019/html/Robinson_Large_Scale_High-Resolution_Land_Cover_Mapping_With_Multi-Resolution_Data_CVPR_2019_paper.html

[23] K. Malkin et al., "Label super-resolution networks," in *Proc. Int. Conf. Learn. Represent.*, 2018. [Online]. Available: https://openreview.net/forum?id=rkxwShA9Ym

[24] R. Dong, W. Fang, H. Fu, L. Gan, J. Wang, and P. Gong, "High-resolution land cover mapping through learning with noise correction," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4402013.

[25] Y. Liu, Y. Zhong, A. Ma, J. Zhao, and L. Zhang, "Cross-resolution national-scale land-cover mapping based on noisy label learning: A case study of China," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 118, Apr. 2023, Art. no. 103265.

[26] Z. Li, H. Wang, Y. Wang, G. Chen, J. M. Chen, and B. Chen, "Large-scale high-resolution essential urban land cover category mapping using a semantic-augmented and noise-tolerant approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, doi: 10.1109/TGRS.2025.3601626.

[27] V. Igwe, B. Salehi, M. Marjani, N. Farhadi, and M. Mahdianpari, "Cost-effective statewide wetland inventory update using weakly supervised deep learning: A case study in Minnesota, USA," *Remote Sens. Appl., Soc. Environ.*, vol. 41, Jan. 2026, Art. no. 101871, doi: 10.1016/j.rsase.2026.101871.

[28] F. Mohammadimanesh, "Large scale land cover mapping in Ontario, Canada, using a deep learning framework," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 17975–17987, 2025. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/10982192/

[29] M. Wang, C. Li, X. Yang, D. Chu, Z. Zhou, and R. Y. K. Lau, "QTU-Net: Quaternion transformer-based U-Net for water body extraction of RGB satellite image," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5634816. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/10595396/

[30] B. Sadeghi, A. A. Alesheikh, A. Jafari, and F. Rezaie, "Performance evaluation of convolutional neural network and vision transformer models for groundwater potential mapping," *J. Hydrol.*, vol. 654, Jun. 2025, Art. no. 132840.

[31] C. Josse et al., "Physical geography and ecosystems in the tropical Andes," in *Climate Change and Biodiversity in the Tropical Andes*. Inter-American Institute for Global Change Research (IAI) and Scientific Committee on Problems of the Environment (SCOPE), 2011, pp. 152–169.

[32] Natural Resources Canada.(2023). *Land Cover, Circa 2020*. [Online]. Available: https://open.canada.ca/data/en/dataset/ee1580ab-a23d-4f86-a09b-79763677eb47

[33] M. Yarmohamadi, A. A. Alesheikh, M. Sharif, and H. Vahidi, "Predicting dust-storm transport pathways using a convolutional neural network and geographic context for impact adaptation and mitigation in urban areas," *Remote Sens.*, vol. 15, no. 9, p. 2468, May 2023.

[34] M. Yarmohamadi, A. A. Alesheikh, and M. Sharif, "Using hybrid deep learning models to predict dust storm pathways with enhanced accuracy," *Climate*, vol. 13, no. 1, p. 16, Jan. 2025.

[35] E. S. Zamanoglu, S. Erbay, E. Cengil, S. Kosunalp, V. Tumen, and K. Demir, "Land cover segmentation using DeepLabV3 and ResNet50," in *Proc. 4th Int. Conf. Commun., Inf., Electron. Energy Syst. (CIEES)*, Nov. 2023, pp. 1–6. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/10378824/

[36] M. Marjani, M. Mahdianpari, S. A. Ahmadi, E. Hemmati, F. Mohammadimanesh, and M. S. Mesgari, "Application of explainable artificial intelligence in predicting wildfire spread: An ASPP-enabled CNN approach," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/10568207/

[37] J. Lu, B. Zhou, B. Wang, and Q. Zhao, "Land cover classification of remote sensing images based on improved DeeplabV3+ network," *J. Phys., Conf. Ser.*, vol. 2400, no. 1, Dec. 2022, Art. no. 012035. [Online]. Available: https://iopscience.iop.org/article/10.1088/1742-6596/2400/1/012035/meta

[38] M. Marjani, S. A. Ahmadi, and M. Mahdianpari, "FirePred: A hybrid multi-temporal convolutional neural network model for wildfire spread prediction," *Ecol. Informat.*, vol. 78, Dec. 2023, Art. no. 102282, doi: 10.1016/j.ecoinf.2023.102282.

[39] M. Sheykhmousa, M. Mahdianpari, H. Ghanbari, F. Mohammadimanesh, P. Ghamisi, and S. Homayouni, "Support vector machine versus random forest for remote sensing image classification: A meta-analysis and systematic review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 6308–6325, 2020.

[40] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[41] E. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Sys. (NIPS)*, vol. 34, Dec. 2021, pp. 12077–12090.

[42] J. Wang et al., "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, Oct. 2021, doi: 10.1109/TPAMI.2020.2983686.

[43] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778. [Online]. Available: http://openaccess.thecvf.com/content_cvpr_2016/html/He_Deep_Residual_Learning_CVPR_2016_paper.html

[44] D. Rocchini et al., "Uncertainty in ecosystem mapping by remote sensing," *Comput. Geosci.*, vol. 50, pp. 128–135, Aug. 2013.

[45] M. Marjani, F. Mohammadimanesh, D. J. Varon, A. Radman, and M. Mahdianpari, "PRISMethaNet: A novel deep learning model for landfill methane detection using PRISMA satellite data," *ISPRS J. Photogramm. Remote Sens.*, vol. 218, pp. 802–818, Dec. 2024.

[46] J. Wang, Y. Zhong, A. Ma, Z. Zheng, Y. Wan, and L. Zhang, "LoveNAS: Towards multi-scene land-cover mapping via hierarchical searching adaptive network," *ISPRS J. Photogramm. Remote Sens.*, vol. 209, pp. 265–278, Mar. 2024, doi: 10.1016/j.isprsjprs.2024.01.011.

[47] A. Midekisa et al., "Mapping land cover change over continental Africa using Landsat and Google Earth engine cloud computing," *PLoS ONE*, vol. 12, no. 9, Sep. 2017, Art. no. e0184926.

[48] M. Marjani, M. Mahdianpari, and F. Mohammadimanesh, "CNN-BiLSTM: A novel deep learning model for near-real-time daily wildfire spread prediction," *Remote Sens.*, vol. 16, no. 8, p. 1467, Apr. 2024.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:44:22 UTC from IEEE Xplore. Restrictions apply.